new idea

WITH

WITH STATEMENT

Find all departments with the maximum budget.

department

dept_name	*building*	*budget*
Biology	Watson	90000.00
Comp. Sci.	Taylor	100000.00
Elec. Eng.	Taylor	85000.00
Finance	Painter	120000.00
History	Painter	50000.00
Music	Packard	80000.00
Physics	Watson	70000.00

```
SELECT dept_name
FROM department
WHERE department.budget IN
          (SELECT max(budget)
           FROM department);
```

dept_name
Finance

an equivalent way

WITH

first create temporary tables, and then use them inside the remaining query.

table name attribute name

```
WITH max_budget (new_attr) AS
       (SELECT max(budget)
        FROM department)

SELECT dept_name
FROM department, max_budget
WHERE department.budget = max_budget.new_attr;
```

WITH STATEMENT

department

dept_name	*building*	*budget*
Biology	Watson	90000.00
Comp. Sci.	Taylor	100000.00
Elec. Eng.	Taylor	85000.00
Finance	Painter	120000.00
History	Painter	50000.00
Music	Packard	80000.00
Physics	Watson	70000.00

Find all departments with the maximum budget.

```
WITH max_budget (new_attr) AS
    (SELECT max(budget)
     FROM department)

SELECT dept_name
FROM department, max_budget
WHERE department.budget = max_budget.new_attr;
```

table name

attribute name

```
WITH max_budget AS
    (SELECT max(budget) as abc
     FROM department)

SELECT dept_name
FROM department, max_budget
WHERE department.budget = max_budget.abc;
```

dept_name
Finance

WITH STATEMENT

```
WITH max_budget (def) AS
    (SELECT max(budget) as abc
     FROM department)

SELECT dept_name
FROM department, max_budget
WHERE department.budget = max_budget.def;
```

dept_name
Finance

```
WITH max_budget (def) AS
    (SELECT max(budget) as abc
     FROM department)

SELECT dept_name
FROM department, max_budget
WHERE department.budget = max_budget.abc;
```

```
myuniversitydb=# WITH max_budget (def) AS
                     (SELECT max(budget) as abc
                      FROM department)

                 SELECT dept_name
                 FROM department, max_budget
                 WHERE department.budget = max_budget.abc;
ERROR:  column max_budget.abc does not exist
LINE 7:                   WHERE department.budget = max_budget.abc;
                                                    ^
```

```
SELECT *
FROM student, takes
WHERE student.id = takes.id;
```

id	name	dept_name	tot_cred	id	course_id	sec_id	semester	year	grade
00128	Zhang	Comp. Sci.	102	00128	CS-347	1	Fall	2009	A-
00128	Zhang	Comp. Sci.	102	00128	CS-101	1	Fall	2009	A
19991	Brandt	History	80	19991	HIS-351	1	Spring	2010	B
23121	Chavez	Finance	110	23121	FIN-201	1	Spring	2010	C+
45678	Levy	Physics	46	45678	CS-319	1	Spring	2010	B
45678	Levy	Physics	46	45678	CS-101	1	Spring	2010	B+
45678	Levy	Physics	46	45678	CS-101	1	Fall	2009	F
54321	Williams	Comp. Sci.	54	54321	CS-190	2	Spring	2009	B+
54321	Williams	Comp. Sci.	54	54321	CS-101	1	Fall	2009	A-
55739	Sanchez	Music	38	55739	MU-199	1	Spring	2010	A-
76543	Brown	Comp. Sci.	58	76543	CS-319	2	Spring	2010	A
76543	Brown	Comp. Sci.	58	76543	CS-101	1	Fall	2009	A
76653	Aoi	Elec. Eng.	60	76653	EE-181	1	Spring	2009	C
98765	Bourikas	Elec. Eng.	98	98765	CS-315	1	Spring	2010	B
98765	Bourikas	Elec. Eng.	98	98765	CS-101	1	Fall	2009	C-
98988	Tanaka	Biology	120	98988	BIO-301	1	Summer	2010	
98988	Tanaka	Biology	120	98988	BIO-101	1	Summer	2009	A
44553	Peltier	Physics	70	44553	PHY-101	1	Fall	2009	B-
12345	Shankar	Comp. Sci.	32	12345	CS-347	1	Fall	2009	A
12345	Shankar	Comp. Sci.	32	12345	CS-315	1	Spring	2010	A
12345	Shankar	Comp. Sci.	32	12345	CS-190	2	Spring	2009	A
12345	Shankar	Comp. Sci.	32	12345	CS-101	1	Fall	2009	C

```
WITH A AS
    (SELECT *
     FROM student, takes
     WHERE student.id = takes.id)

SELECT *
FROM A;
```

id	name	dept_name	tot_cred	id	course_id	sec_id	semester	year	grade
00128	Zhang	Comp. Sci.	102	00128	CS-347	1	Fall	2009	A-
00128	Zhang	Comp. Sci.	102	00128	CS-101	1	Fall	2009	A
19991	Brandt	History	80	19991	HIS-351	1	Spring	2010	B
23121	Chavez	Finance	110	23121	FIN-201	1	Spring	2010	C+
45678	Levy	Physics	46	45678	CS-319	1	Spring	2010	B
45678	Levy	Physics	46	45678	CS-101	1	Spring	2010	B+
45678	Levy	Physics	46	45678	CS-101	1	Fall	2009	F
54321	Williams	Comp. Sci.	54	54321	CS-190	2	Spring	2009	B+
54321	Williams	Comp. Sci.	54	54321	CS-101	1	Fall	2009	A-
55739	Sanchez	Music	38	55739	MU-199	1	Spring	2010	A-
76543	Brown	Comp. Sci.	58	76543	CS-319	2	Spring	2010	A
76543	Brown	Comp. Sci.	58	76543	CS-101	1	Fall	2009	A
76653	Aoi	Elec. Eng.	60	76653	EE-181	1	Spring	2009	C
98765	Bourikas	Elec. Eng.	98	98765	CS-315	1	Spring	2010	B
98765	Bourikas	Elec. Eng.	98	98765	CS-101	1	Fall	2009	C-
98988	Tanaka	Biology	120	98988	BIO-301	1	Summer	2010	
98988	Tanaka	Biology	120	98988	BIO-101	1	Summer	2009	A
44553	Peltier	Physics	70	44553	PHY-101	1	Fall	2009	B-
12345	Shankar	Comp. Sci.	32	12345	CS-347	1	Fall	2009	A
12345	Shankar	Comp. Sci.	32	12345	CS-315	1	Spring	2010	A
12345	Shankar	Comp. Sci.	32	12345	CS-190	2	Spring	2009	A
12345	Shankar	Comp. Sci.	32	12345	CS-101	1	Fall	2009	C

WITH STATEMENT

teacher

id	name	dept_name	salary
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

can create several tables in WITH

Find all departments where teachers' total salary is at least the average of the total salary of all departments.

```
WITH dept_total (dept_name, value) AS
        (SELECT dept_name, sum(salary)
         FROM teacher
         GROUP BY dept_name),
     dept_total_avg (value) AS
        (SELECT avg(value)
         FROM dept_total)
SELECT dept_name
FROM dept_total, dept_total_avg
WHERE dept_total.value >= dept_total_avg.value
```

dept_total

dept_name	value
Finance	170000.00
History	122000.00
Physics	182000.00
Music	40000.00
Comp. Sci.	232000.00
Biology	72000.00
Elec. Eng.	80000.00

WITH STATEMENT

teacher

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

Find all departments where teachers' total salary is at least the average of the total salary of all departments.

```
WITH dept_total (dept_name, value) AS
        (SELECT dept_name, sum(salary)
        FROM teacher
        GROUP BY dept_name),
    dept_total_avg (value) AS
        (SELECT avg(value)
        FROM dept_total)
SELECT dept_name
FROM dept_total, dept_total_avg
WHERE dept_total.value >= dept_total_avg.value
```

dept_total

dept_name	*value*
Finance	170000.00
History	122000.00
Physics	182000.00
Music	40000.00
Comp. Sci.	232000.00
Biology	72000.00
Elec. Eng.	80000.00

value
128285.714285714286

dept_total_avg

WITH STATEMENT

teacher

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

Find all departments where teachers' total salary is at least the average of the total salary of all departments.


```
WITH dept_total (dept_name, value) AS
    (SELECT dept_name, sum(salary)
     FROM teacher
     GROUP BY dept_name),

  dept_total_avg (value) AS
    (SELECT avg(value)
     FROM dept_total)

SELECT dept_name
FROM dept_total, dept_total_avg
WHERE dept_total.value >= dept_total_avg.value;
```


dept_total

dept_name	*value*
Finance	170000.00
History	122000.00
Physics	182000.00
Music	40000.00
Comp. Sci.	232000.00
Biology	72000.00
Elec. Eng.	80000.00

value
128285.714285714286

dept_total_avg

dept_name
Finance
Physics
Comp. Sci.

WITH STATEMENT

Find all departments where teachers' total salary is at least the average of the total salary of all departments.

```
WITH dept_total (dept_name, value) AS
    (SELECT dept_name, sum(salary)
     FROM teacher
     GROUP BY dept_name),

dept_total_avg (value) AS
    (SELECT avg(value)
     FROM dept_total)

SELECT dept_name
FROM dept_total, dept_total_avg
WHERE dept_total.value >= dept_total_avg.value;
```

cannot shorten to `dept_total_avg`

WITH

can always be replaced with nested queries but WITH is more logical, linear, easier.

CREATE TEMPORARY TABLE

RECALL: create a temporary table immediately inside a query:

```
SELECT *
FROM ( VALUES ('a',10), ('b', 20) ) AS A(f,s);
```

A

f	*s*
a	10
b	20

WITH can also be used to create a temporary table

```
WITH T(f,s) AS
 (SELECT *
  FROM ( VALUES ('john',10),
                ('james', 16),
                ('joe', 20) )
 )

SELECT *
FROM T;
```

T

f	*s*
john	10
james	16
joe	20

```
WITH mynewtable (name, grade) AS
    (SELECT 'john', 10
      UNION ALL
     SELECT 'james', 16
      UNION ALL
     SELECT 'joe', 20)

SELECT *
FROM mynewtable;
```

name	*grade*
john	10
james	16
joe	20

mynewtable

new idea

SET OPERATIONS

UNION

`UNION` of sets $\{a, b, e\} \bigcup \{b, m, n\} = \{a, b, e, m, n\}$

as a table is just a set of tuples, we can do the same for tables !

automatically removes all duplicate tuples!

(**SELECT** course_id
FROM section
WHERE semester = 'Spring' **AND** year = 2010) →

course_id
CS-101
CS-315
CS-319
CS-319
FIN-201
HIS-351
MU-199

UNION →

course_id
CS-101
CS-315
CS-319
CS-347
FIN-201
HIS-351
MU-199
PHY-101

(**SELECT** course_id
FROM section
WHERE semester = 'Fall' **AND** year = 2009) →

course_id
CS-101
CS-347
PHY-101

UNION

`UNION` of sets $\{a, b, e\} \bigcup \{b, m, n\} = \{a, b, e, m, n\}$

as a table is just a set of tuples, we can do the same for tables !

to keep duplicates, use UNION ALL

(**SELECT** course_id
FROM section
WHERE semester = 'Spring' **AND** year = 2010)

course_id
CS-101
CS-315
CS-319
CS-319
FIN-201
HIS-351
MU-199

UNION ALL

course_id
CS-101
CS-315
CS-319
CS-319
FIN-201
HIS-351
MU-199
CS-101
CS-347
PHY-101

(**SELECT** course_id
FROM section
WHERE semester = 'Fall' **AND** year = 2009)

course_id
CS-101
CS-347
PHY-101

attr1
a
a
b

UNION

attrib2
a
b

→

attr1
a
b

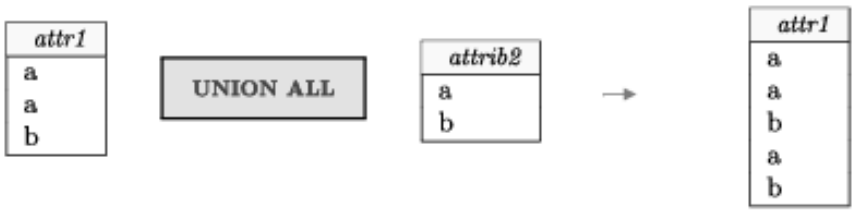
```
(SELECT *
 FROM (VALUES ('a'), ('a'), ('b')) AS table1(attr1))

   UNION

(SELECT *
 FROM (VALUES ('a'), ('b')) AS table2(attr2));
```

↓

attr1
b
a

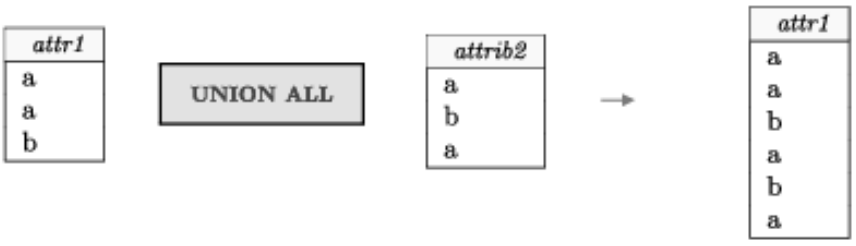



```
(SELECT *
 FROM (VALUES ('a'), ('a'), ('b')) AS table1(attr1))

   UNION ALL

(SELECT *
 FROM (VALUES ('a'), ('b')) AS table2(attr2));
```

attr1
a
a
b
a
b




```
(SELECT *
 FROM (VALUES ('a'), ('a'), ('b')) AS table1(attr1))

   UNION ALL

(SELECT *
 FROM (VALUES ('a'), ('b'), ('a')) AS table2(attr2));
```

attr1
a
a
b
a
b
a

Set Operations

INTERSECT sets $\{a, b, e\} \bigcap \{b, m, n\} = \{b\}$

as a table is just a set of tuples, we can do the same for tables !

automatically removes all duplicate tuples!

```
(SELECT course_id
 FROM section
 WHERE semester = 'Spring' AND
       year = 2010)
```

course_id
CS-101
CS-315
CS-319
CS-319
FIN-201
HIS-351
MU-199

INTERSECT

course_id
CS-101

```
(SELECT course_id
 FROM section
 WHERE semester = 'Fall' AND
       year = 2009)
```

course_id
CS-101
CS-347
PHY-101

INTERSECT ALL

keep minimum of duplicates in both tables

attr1
a
a
b

INTERSECT

attrib2
a
b

→

attr1
a
b

```
(SELECT *
 FROM (VALUES ('a'), ('a'), ('b')) AS table1(attr1))

    INTERSECT

(SELECT *
 FROM (VALUES ('a'), ('b')) AS table2(attr2));
```

↓

attr1
b
a

attr1
a
a
b

INTERSECT ALL

attrib2
a
b
a

→

attr1
b
a
a

```
(SELECT *
 FROM (VALUES ('a'), ('a'), ('b')) AS table1(attr1))

    INTERSECT ALL

(SELECT *
 FROM (VALUES ('a'), ('b'), ('a')) AS table2(attr2));
```

↓

attr1
b
a
a

Set Operations

EXCEPT $\{a, b, e\} \setminus \{b, m, n\} = \{a, e\}$

as a table is just a set of tuples, we can do the same for tables !

(**SELECT** course_id
FROM section
WHERE semester = 'Spring' **AND**
year = 2010)

→

course_id
CS-101
CS-315
CS-319
CS-319
FIN-201
HIS-351
MU-199

automatically removes all duplicate tuples!

EXCEPT →

course_id
HIS-351
CS-319
FIN-201
CS-315
MU-199

(**SELECT** course_id
FROM section
WHERE semester = 'Fall' **AND**
year = 2009)

→

course_id
CS-101
CS-347
PHY-101

Set Operations

EXCEPT $\{a, b, e\} \setminus \{b, m, n\} = \{a, e\}$

as a table is just a set of tuples, we can do the same for tables !

to keep duplicates, use EXCEPT ALL

(**SELECT** course_id
FROM section
WHERE semester = 'Spring' **AND** year = 2010)

course_id
CS-101
CS-315
CS-319
CS-319
FIN-201
HIS-351
MU-199

EXCEPT ALL

course_id
HIS-351
CS-319
CS-319
FIN-201
CS-315
MU-199

(**SELECT** course_id
FROM section
WHERE semester = 'Fall' **AND** year = 2009)

course_id
CS-101
CS-347
PHY-101

attr1
a
a
a
b

EXCEPT ALL

attrib2
a
b
a

→ *attr1*

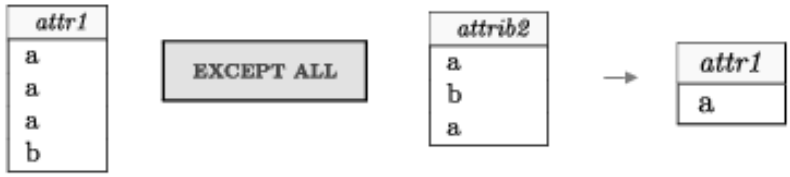
```
(SELECT *
 FROM (VALUES ('a'), ('a'), ('a'), ('b')) AS table1(attr1))

    EXCEPT

(SELECT *
 FROM (VALUES ('a'), ('b'), ('a')) AS table2(attr2));
```

↓

attr1



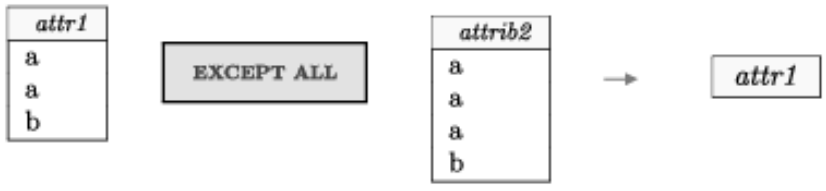

```
(SELECT *
 FROM (VALUES ('a'), ('a'), ('a'), ('b')) AS table1(attr1))

   EXCEPT ALL

(SELECT *
 FROM (VALUES ('a'), ('b'), ('a')) AS table2(attr2));
```



attr1
a




```
(SELECT *
 FROM (VALUES ('a'), ('a'), ('b')) AS table1(attr1))

   EXCEPT ALL

(SELECT *
 FROM (VALUES ('a'), ('b'), ('a')) AS table2(attr2));
```



attr1

number of attributes must be the same

```
myuniversitydb=# (SELECT dept_name FROM teacher)
myuniversitydb-#    EXCEPT
myuniversitydb-# (SELECT name, salary FROM teacher);
ERROR:  each EXCEPT query must have the same number of columns
LINE 3: (SELECT name, salary FROM teacher);
                ^
```

domain types of attributes must be the same

```
myuniversitydb=# (SELECT dept name, id FROM teacher)
                     EXCEPT
                (SELECT name, salary FROM teacher);
ERROR:  EXCEPT types character varying and numeric cannot be matched
LINE 3:                  (SELECT name, salary FROM teacher);
                                       ^
```

attribute names need *not* match

```
(SELECT dept_name FROM teacher)
   EXCEPT
(SELECT name FROM teacher);
```

→

dept_name
Physics
Biology
Elec. Eng.
Finance
Comp. Sci.
History
Music

new idea

STRINGS

Strings

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

`varchar` (N): variable-length string of N characters

Strings are represented by single quotes: 'Comp. Sci.'

~~"Comp. Sci."~~

Pattern matching with the keyword `LIKE`

% : any sequence of characters (possibly empty)

```
WHERE name LIKE '%an%'
```

any string containing 'an' as a substring

name
Srinivasan
Brandt

_ : single character matching

```
WHERE name LIKE '__an%'
```

any string containing 'an' after two chars

name
Brandt

many string functions available, depends on database system

Strings

`LIKE '%an%'`	containing 'an' as a substring
`LIKE 'an%'`	starting with 'an'
`LIKE '%an'`	ending with 'an'
`LIKE 'wh%an'`	starting with 'wh', ending with 'an'
`LIKE 'a_n'`	any single character between 'a' and 'n'

STRINGS

teacher

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

pattern matching only works with LIKE

```
universitydb=# SELECT *
               FROM teacher
               WHERE dept name = '%o%';
 id | name | dept_name | salary
----+------+-----------+--------
(0 rows)
```

```
universitydb=# SELECT *
               FROM teacher
               WHERE dept name LIKE '%o%';
  id   |    name    | dept name  |  salary
-------+------------+------------+----------
 10101 | Srinivasan | Comp. Sci. | 65000.00
 32343 | El Said    | History    | 60000.00
 45565 | Katz       | Comp. Sci. | 75000.00
 58583 | Califieri  | History    | 62000.00
 76766 | Crick      | Biology    | 72000.00
 83821 | Brandt     | Comp. Sci. | 92000.00
(6 rows)
```

```
universitydb=# SELECT *
               FROM teacher
               WHERE dep. name LIKE 'Comp. Sci.';
  id   |    name    | dept name  |  salary
-------+------------+------------+----------
 10101 | Srinivasan | Comp. Sci. | 65000.00
 45565 | Katz       | Comp. Sci. | 75000.00
 83821 | Brandt     | Comp. Sci. | 92000.00
(3 rows)
```

```
universitydb=# SELECT *
universitydb-# FROM teacher
universitydb-# WHERE dept name = 'Comp. Sci.';
  id   |    name    | dept name  |  salary
-------+------------+------------+----------
 10101 | Srinivasan | Comp. Sci. | 65000.00
 45565 | Katz       | Comp. Sci. | 75000.00
 83821 | Brandt     | Comp. Sci. | 92000.00
(3 rows)
```

STRINGS

Function	Return type	Description	Example	Result
[illegible]	text	String concatenation	[illegible]	[illegible]
[illegible]	text	string concatenation with one non-string input	[illegible]	[illegible]
[illegible]	int	Number of bits in string	[illegible]	[illegible]
[illegible]	int	Number of characters in string	[illegible]	[illegible]
[illegible]	text	Convert string to lower case	[illegible]	[illegible]
[illegible]	int	Number of bytes in string	[illegible]	[illegible]
[illegible]	text	Replace substring	[illegible]	[illegible]

https://www.postgresql.org/docs/9.1/functions-string.html

new idea

NULL

`null` : indicates **missing, unknown** or **incomplete** data

Rules in dealing with null

- result of any arithmetic operation involving null ⟶ null

`2 + null → null`

```
myuniversitydb=# SELECT salary+null AS newsalary
myuniversitydb-# FROM teacher;
 newsalary
-----------
(12 rows)
```

the result of a comparison can be **true** or **false** or **null**

```
SELECT T1.id, T1.grade, T2.id, T2.grade, (T1.grade = T2.grade)
FROM takes as T1, takes as T2
WHERE T1.id = '98988' AND T1.year = 2010;
```

takes

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

id	*grade*	*id*	*grade*	*?column?*
98988		00128	A	
98988		00128	A-	
98988		12345	C	
98988		12345	A	
98988		12345	A	
98988		12345	A	
98988		19991	B	
98988		23121	C+	
98988		44553	B-	
98988		45678	F	
98988		45678	B+	
98988		45678	B	
98988		54321	A-	
98988		54321	B+	
98988		55739	A-	
98988		76543	A	
98988		76543	A	
98988		76653	C	
98988		98765	C-	
98988		98765	B	
98988		98988	A	
98988		98988		

`null` : indicates missing, unknown or incomplete data

Rules in dealing with null

- **WHERE** clause evaluates to null or **false** $\longrightarrow$ row not added
- a row ('abc', null) is considered equal to another row ('abc', null) for:

union, intersect, except

in UNION / INTERSECT / EXCEPT, two NULLs count as **equal** — unlike = comparisons!

```
(SELECT 'abc', null AS a)
   EXCEPT
(SELECT 'abc', null AS b);
```

?column?	*a*

```
(SELECT 'abc', null AS a)
   UNION
(SELECT 'abc', null AS b);
```

?column?	*a*
abc	

`null` : indicates missing, unknown or incomplete data

Rules in dealing with null

- aggregate functions ignore null values

takes

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

```
SELECT count(*)
FROM takes;
```

count
22

```
SELECT count(grade)
FROM takes;
```

count
21

NULL

`null` : indicates missing, unknown or incomplete data

Comparisons

true **and** null ⟶ null
false **and** null ⟶ false
null **and** null ⟶ null

true **or** null ⟶ true
false **or** null ⟶ null
null **or** null ⟶ null

not null ⟶ null

```
SELECT NOT NULL;
```

?column?

```
SELECT TRUE OR NULL;
```

?column?
t

```
SELECT 'hello' IN (null, 'hello');
```

?column?
t

```
SELECT 'hello' NOT IN (null, 'hello');
```

?column?
f

```
SELECT 'hello' IN (null, 'bye');
```

?column?

```
SELECT 'hello' NOT IN (null, 'bye');
```

?column?

How to check if an attribute `attr` is null ?

(`attr` = null) is never true ✗

```
SELECT * FROM takes WHERE grade = NULL;
```

id	course_id	sec_id	semester	year	grade

(`attr` ! = null) is never true ✗

```
SELECT * FROM takes WHERE grade != NULL;
```

id	course_id	sec_id	semester	year	grade

must use: (`attr IS null`) ✓

```
SELECT * FROM takes WHERE grade IS NULL;
```

id	course_id	sec_id	semester	year	grade
98988	BIO-301	1	Summer	2010	

```
SELECT *
FROM teacher
WHERE salary > NULL;
```

(salary > null) is never true

✗

id	name	dept_name	salary	specialties

(salary ! = null) is never true

```
SELECT *
FROM teacher
WHERE salary != NULL;
```

✗

id	name	dept_name	salary	specialties

(salary < null) is never true

```
SELECT *
FROM teacher
WHERE salary < NULL;
```

✗

id	name	dept_name	salary	specialties

NULL

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

```
SELECT (grade IS null)
FROM takes;
```

?column?
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
f
t

In psql, empty cell in an output table means NULL.

```
SELECT *
FROM takes
WHERE grade LIKE 'A'
      OR grade IS NULL;
```

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

```
myuniversitydb=# \pset null 'NULL'
Null display is "NULL".
```

```
SELECT *
FROM takes
WHERE grade LIKE 'A'
      OR grade IS NULL;
```

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	NULL

(`attr` = null) is never true

(`attr` ! = null) is never true

Find all courses where did not get 'A' grade.

takes

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

```
SELECT *
FROM takes
WHERE grade NOT LIKE 'A';
```

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B

missing null row... ✗

(attr = null) is never true

(attr ! = null) is never true

Find all courses where did not get 'A' grade.

takes

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

```
SELECT *
FROM takes
WHERE grade NOT LIKE 'A'
      OR grade IS NULL;
```

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-301	1	Summer	2010	

✓

```
SELECT *
FROM takes
ORDER BY grade NULLS FIRST;
```

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
98988	BIO-301	1	Summer	2010	
00128	CS-101	1	Fall	2009	A
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
98988	BIO-101	1	Summer	2009	A
00128	CS-347	1	Fall	2009	A-
54321	CS-101	1	Fall	2009	A-
55739	MU-199	1	Spring	2010	A-
45678	CS-319	1	Spring	2010	B
98765	CS-315	1	Spring	2010	B
19991	HIS-351	1	Spring	2010	B
54321	CS-190	2	Spring	2009	B+
45678	CS-101	1	Spring	2010	B+
44553	PHY-101	1	Fall	2009	B-
12345	CS-101	1	Fall	2009	C
76653	EE-181	1	Spring	2009	C
23121	FIN-201	1	Spring	2010	C+
98765	CS-101	1	Fall	2009	C-
45678	CS-101	1	Fall	2009	F

```
SELECT *
FROM takes
ORDER BY grade NULLS LAST;
```

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
76543	CS-319	2	Spring	2010	A
76543	CS-101	1	Fall	2009	A
98988	BIO-101	1	Summer	2009	A
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
00128	CS-101	1	Fall	2009	A
55739	MU-199	1	Spring	2010	A-
00128	CS-347	1	Fall	2009	A-
54321	CS-101	1	Fall	2009	A-
45678	CS-319	1	Spring	2010	B
19991	HIS-351	1	Spring	2010	B
98765	CS-315	1	Spring	2010	B
54321	CS-190	2	Spring	2009	B+
45678	CS-101	1	Spring	2010	B+
44553	PHY-101	1	Fall	2009	B-
12345	CS-101	1	Fall	2009	C
76653	EE-181	1	Spring	2009	C
23121	FIN-201	1	Spring	2010	C+
98765	CS-101	1	Fall	2009	C-
45678	CS-101	1	Fall	2009	F
98988	BIO-301	1	Summer	2010	

new idea

ALL, SOME, EXISTS

ALL

teacher

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

Find the names of teachers with salary greater than **all** teachers in 'Comp. Sci.' department.

```
SELECT name
FROM teacher
WHERE salary > ALL (SELECT T.salary
                    FROM teacher AS T
                    WHERE T.dept_name = 'Comp. Sci.');
```

ALL

true if *all* elements of the list satisfy the condition

name
Einstein

ALL

teacher

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

```
SELECT *
FROM teacher
WHERE salary >= ALL ( VALUES (70000),(90000) );
```



id	*name*	*dept_name*	*salary*
12121	Wu	Finance	90000.00
22222	Einstein	Physics	95000.00
83821	Brandt	Comp. Sci.	92000.00

```
SELECT *
FROM teacher
WHERE salary >= ALL ( VALUES (120000),(90000) );
```



id	*name*	*dept_name*	*salary*

ALL

teacher

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

Find the department names and number of teachers in that department, but only for those departments with **maximum number of teachers.**

```
SELECT A.dept_name, count(*)
FROM teacher as A
GROUP BY A.dept_name
HAVING count(*) >= ( SELECT max(c)
                     FROM (SELECT count(*) as c
                           FROM teacher as B
                           GROUP BY B.dept_name)
                   );
```

with ALL

```
SELECT dept_name, count(*)
FROM teacher
GROUP BY dept_name
HAVING count(*) >= ALL(SELECT count(*)
                       FROM teacher
                       GROUP BY dept_name);
```


dept_name	*count*
Comp. Sci.	3

SOME

teacher

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

Find the names of teachers with salary greater than **some** teacher in 'Comp. Sci.' department.

```
SELECT name
FROM teacher
WHERE salary >
      SOME (SELECT T. salary
            FROM teacher AS T
            WHERE T.dept_name = 'Comp. Sci.');
```

SOME

true if *at least one tuple* of the list satisfies the condition

Equivalent keyword: **ANY**

name
Wu
Einstein
Gold
Katz
Singh
Crick
Brandt
Kim

SOME

teacher

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

```
SELECT *
FROM teacher
WHERE salary >= SOME ( VALUES (70000),(90000) );
```

id	*name*	*dept_name*	*salary*
76543	Singh	Finance	80000.00
98345	Kim	Elec. Eng.	80000.00
12121	Wu	Finance	90000.00
22222	Einstein	Physics	95000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00

```
SELECT *
FROM teacher
WHERE salary >= SOME ( VALUES (120000),(90000) );
```

id	*name*	*dept_name*	*salary*
12121	Wu	Finance	90000.00
22222	Einstein	Physics	95000.00
83821	Brandt	Comp. Sci.	92000.00

EXAMPLE

takes

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

student

id	*name*	*dept_name*	*tot_cred*
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

Find the name of the student who has taken the most classes.

Step 1: create table with number of classes taken by each student id

```
SELECT id, count(*) as mynum
FROM takes
GROUP BY id
```

id	*mynum*
76653	1
19991	1
76543	2
54321	2
44553	1
55739	1
45678	3
12345	4
98988	2
98765	2
00128	2
23121	1

Step 2: pick max from this table

```
SELECT max(mynum)
FROM ( SELECT id, count(*) as mynum
       FROM takes
       GROUP BY id ) as mytable
```

max
4

takes

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

student

id	*name*	*dept_name*	*tot_cred*
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

Step 3: find the student id corresponding to this max value

```
SELECT id
FROM takes
GROUP BY id
HAVING count(*) = (
      SELECT max(mynum)
      FROM ( SELECT count(*) as mynum
             FROM takes
             GROUP BY id ) as mytable
      )
```

id
12345

Step 4: product with student table to find the student name

```
SELECT student.name
FROM student, takes
WHERE student.id = takes.id
GROUP BY student.name
HAVING count(*) =
            (SELECT max(mynum)
             FROM (SELECT count(*) as mynum
                   FROM takes
                   GROUP BY id ) as mytable
            );
```

name
Shankar

EXAMPLE

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

id	name	dept_name	tot_cred
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

student

Find the name of the student who has taken the most classes.

Step 1: create table with number of classes taken by each student id

```
SELECT id, count(*) as mynum
FROM takes
GROUP BY id
```

id	mynum
76653	1
19991	1
76543	2
54321	2
44553	1
55739	1
45678	3
12345	4
98988	2
98765	2
00128	2
23121	1

Step 2: pick student id which appears the most using **ALL**

```
SELECT id
FROM takes
GROUP BY id
HAVING count(*) >= ALL (
                SELECT count(*)
                FROM takes
                GROUP BY id )
```

id
12345

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

id	name	dept_name	tot_cred
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

student

Step 3: product with the student table to find the student name

```
SELECT student.name
FROM takes, student
WHERE takes.id = student.id
GROUP BY student.name
HAVING count(*) >= ALL (SELECT count(*)
                        FROM takes
                        GROUP BY id);
```

name
Shankar

EXISTS

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

Find the list of students who have taken a class in 2009.

```
SELECT S.id, S.name
FROM student as S
WHERE EXISTS (SELECT *
              FROM takes as T
              WHERE T.id = S.id
                    AND T.year = 2009);
```

id	name	dept_name	tot_cred
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

student

EXISTS (input_table)

true if input_table is non-empty

id	name
00128	Zhang
45678	Levy
54321	Williams
76543	Brown
76653	Aoi
98765	Bourikas
98988	Tanaka
44553	Peltier
12345	Shankar

EXISTS

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

Find the list of students who have not taken any class in 2009.

```
SELECT S.id, S.name
FROM student as S
WHERE NOT EXISTS (SELECT *
                  FROM takes as T
                  WHERE T.id = S.id
                        AND T.year = 2009);
```

id	name	dept_name	tot_cred
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

student

NOT EXISTS (input_table)

true if input_table is empty

id	name
19991	Brandt
23121	Chavez
55739	Sanchez
70557	Snow

EXISTS

```
SELECT EXISTS (SELECT *
               FROM teacher
               WHERE salary < 10000);
```

exists
f

```
SELECT NOT EXISTS (SELECT *
                   FROM teacher
                   WHERE salary < 10000);
```

?column?
t

Why is the last one `true`?
EXISTS only checks whether the subquery returns **any row**: `SELECT count(*)` always returns **one** row (here with value 0), so it is **`true`**.

```
SELECT EXISTS (SELECT count(*)
               FROM teacher
               WHERE salary < 10000);
```

exists
t

EXISTS

takes

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

student

id	*name*	*dept_name*	*tot_cred*
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

For each student, print **true** or **false** if they have an 'A' grade for some course.

```
SELECT S.name, EXISTS (SELECT *
                       FROM takes AS T
                       WHERE S.id = T.id AND T.grade = 'A')
FROM student as S;
```

name	*exists*
Zhang	t
Shankar	t
Brandt	f
Chavez	f
Peltier	f
Levy	f
Williams	f
Sanchez	f
Snow	f
Brown	t
Aoi	f
Bourikas	f
Tanaka	t

takes

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

EXISTS

Find the list of students who have not taken any class in 2009.

```
SELECT S.id, S.name
FROM student as S
WHERE NOT EXISTS (SELECT *
                  FROM takes as T
                  WHERE T.id = S.id
                        AND T.year = 2009);
```

id	*name*
19991	Brandt
23121	Chavez
55739	Sanchez
70557	Snow

id	*name*	*dept_name*	*tot_cred*
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

student

called correlated subquery

Very Slow

note that inner query uses outer query table S

inner query run once for *each* tuple of S

generally very slow, should find alternative ways to get the same result

takes

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

EXAMPLE

Find the list of students who have not taken any class in 2009.

```
(SELECT student.id, student.name
 FROM student)

      EXCEPT

(SELECT student.id, student.name
 FROM takes, student
 WHERE takes.id = student.id and takes.year = 2009);
```

id	*name*	*dept_name*	*tot_cred*
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

student

id	*name*
55739	Sanchez
19991	Brandt
23121	Chavez
70557	Snow

new idea

CASE

Case statement

teacher

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

```
SELECT id, name, dept_name, (CASE dept_name
            WHEN 'Comp. Sci.' THEN 'CS'
            WHEN 'History' THEN 'Histoire'
            ELSE dept_name END)
FROM teacher;
```

→

id	*name*	*dept_name*	*dept_name*
10101	Srinivasan	Comp. Sci.	CS
12121	Wu	Finance	Finance
15151	Mozart	Music	Music
22222	Einstein	Physics	Physics
32343	El Said	History	Histoire
33456	Gold	Physics	Physics
45565	Katz	Comp. Sci.	CS
58583	Califieri	History	Histoire
76543	Singh	Finance	Finance
76766	Crick	Biology	Biology
83821	Brandt	Comp. Sci.	CS
98345	Kim	Elec. Eng.	Elec. Eng.

```
SELECT id, name, dept_name, (CASE
            WHEN dept_name = 'Comp. Sci.' THEN 'CS'
            WHEN dept_name = 'History' THEN 'Histoire'
            ELSE dept_name END) as new_name
FROM teacher;
```

→

id	*name*	*dept_name*	*new_name*
10101	Srinivasan	Comp. Sci.	CS
12121	Wu	Finance	Finance
15151	Mozart	Music	Music
22222	Einstein	Physics	Physics
32343	El Said	History	Histoire
33456	Gold	Physics	Physics
45565	Katz	Comp. Sci.	CS
58583	Califieri	History	Histoire
76543	Singh	Finance	Finance
76766	Crick	Biology	Biology
83821	Brandt	Comp. Sci.	CS
98345	Kim	Elec. Eng.	Elec. Eng.

Case statement

can combine with **aggregations**

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

```
SELECT id, AVG (CASE
            WHEN grade = 'A' THEN 4.0
            WHEN grade = 'B' THEN 3.0
            WHEN grade = 'C' THEN 2.0
            WHEN grade = 'D' THEN 1.0
            ELSE 0.0 END)
FROM takes
GROUP BY id;
```


id	avg
76653	2.0000000000000000
19991	3.0000000000000000
76543	4.0000000000000000
54321	0.00000000000000000000
44553	0.00000000000000000000
55739	0.00000000000000000000
45678	1.00000000000000000000
12345	3.5000000000000000
98988	2.0000000000000000
98765	1.5000000000000000
00128	2.0000000000000000
23121	0.00000000000000000000

```
myuniversitydb=# SELECT id, AVG (CASE
                                 WHEN grade = 'A' THEN 4.0
                                 WHEN grade = 'B' THEN 3.0
                                 WHEN grade = 'C' THEN 2.0
                                 WHEN grade = 'D' THEN 1.0
                                 ELSE 0.0 END)
                FROM takes
                GROUP BY id;
  id   |           avg
-------+------------------------
 76653 |     2.0000000000000000
 19991 |     3.0000000000000000
 76543 |     4.0000000000000000
 54321 | 0.00000000000000000000
 44553 | 0.00000000000000000000
 55739 | 0.00000000000000000000
 45678 | 1.00000000000000000000
 12345 |     3.5000000000000000
 98988 |     2.0000000000000000
 98765 |     1.5000000000000000
 00128 |     2.0000000000000000
 23121 | 0.00000000000000000000
(12 rows)
```

Case statement

can combine with **aggregations**

```
SELECT
  COUNT(CASE WHEN salary > 80000 THEN 1 END) as high,
  COUNT(CASE WHEN salary BETWEEN 50000 AND 80000 THEN 1 END) as medium,
  COUNT(CASE WHEN salary < 50000 THEN 1 END) as low
FROM teacher;
```

teacher

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

→

high	*medium*	*low*
4	7	1

can use CASE in **ORDER BY**

Print table `teacher` with 'Comp. Sci.' before everyone, then sorted by salary

```
SELECT *
FROM teacher
ORDER BY (CASE
          WHEN dept_name = 'Comp. Sci.' THEN 1
          ELSE 2 END), salary;
```

↓

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
45565	Katz	Comp. Sci.	75000.00
83821	Brandt	Comp. Sci.	92000.00
15151	Mozart	Music	40000.00
32343	El Said	History	60000.00
58583	Califieri	History	62000.00
76766	Crick	Biology	72000.00
76543	Singh	Finance	80000.00
98345	Kim	Elec. Eng.	80000.00
33456	Gold	Physics	87000.00
12121	Wu	Finance	90000.00
22222	Einstein	Physics	95000.00

```
SELECT s.ID, s.name, s.dept_name, s.tot_cred, (CASE
            WHEN s.tot_cred >= 100 THEN 'Senior'
            WHEN s.tot_cred >= 60 THEN 'Junior'
            WHEN s.tot_cred >= 30 THEN 'Sophomore'
            ELSE 'Freshman' END) AS standing,
       (CASE WHEN EXISTS (SELECT 1 FROM takes t WHERE t.ID = s.ID
                                        AND (t.grade = 'F' OR t.grade IS NULL)) THEN 'At Risk'
             WHEN s.tot_cred < (SELECT AVG(tot_cred) FROM student) THEN 'Below Average Credits'
             ELSE 'Good Standing' END) AS status,
       (CASE WHEN (SELECT COUNT(*) FROM takes t WHERE t.ID = s.ID AND t.grade IN ('A', 'A-')) >
                  (SELECT COUNT(*) FROM takes t WHERE t.ID = s.ID
                                        AND t.grade IN ('C', 'C-', 'D', 'F')) THEN 'High Achiever'
             WHEN (SELECT COUNT(*) FROM takes t WHERE t.ID = s.ID
                                        AND t.grade = 'F') > 2 THEN 'Academic Probation'
             ELSE 'Standard' END) AS category
FROM student s;
```

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

id	name	dept_name	tot_cred
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

can combine with **subqueries**

id	*name*	*dept_name*	*tot_cred*	*standing*	*status*	*category*
00128	Zhang	Comp. Sci.	102	Senior	Good Standing	High Achiever
12345	Shankar	Comp. Sci.	32	Sophomore	Below Average Credits	High Achiever
19991	Brandt	History	80	Junior	Good Standing	Standard
23121	Chavez	Finance	110	Senior	Good Standing	Standard
44553	Peltier	Physics	56	Sophomore	Below Average Credits	Standard
45678	Levy	Physics	46	Sophomore	At Risk	Standard
54321	Williams	Comp. Sci.	54	Sophomore	Below Average Credits	High Achiever
55739	Sanchez	Music	38	Sophomore	Below Average Credits	High Achiever
70557	Snow	Physics	0	Freshman	Below Average Credits	Standard
76543	Brown	Comp. Sci.	58	Sophomore	Below Average Credits	High Achiever
76653	Aoi	Elec. Eng.	60	Junior	Below Average Credits	Standard
98765	Bourikas	Elec. Eng.	98	Junior	Good Standing	Standard
98988	Tanaka	Biology	120	Senior	At Risk	High Achiever

SQL Style Guidelines

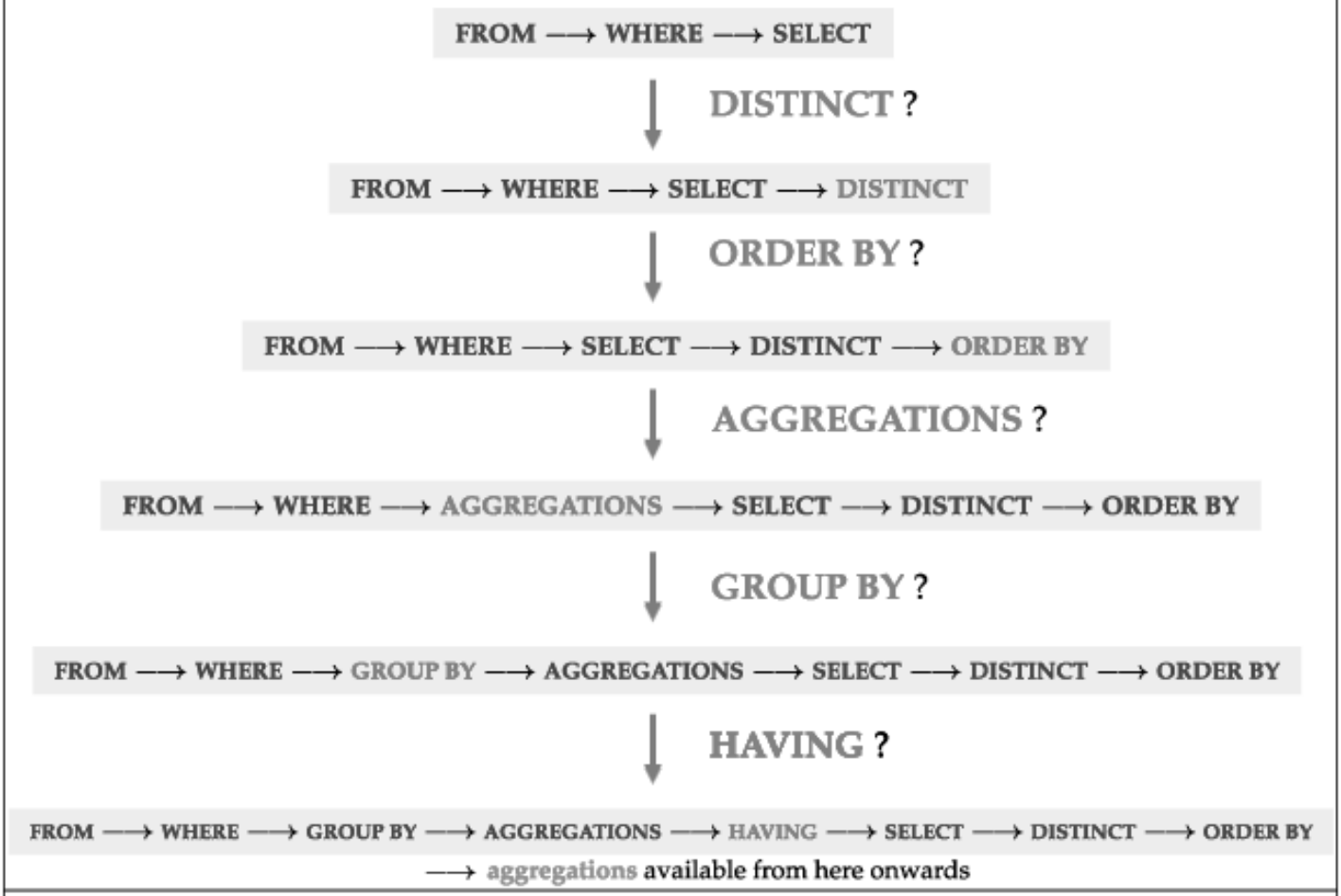
use lower-case names for tables, attributes, etc.

use capital case for SQL keywords (**FROM**, . . .)

follow uniform indentation scheme

don't have too long lines (try for 80 characters or less)

Logical Order of Processing



new idea

JOINS

PRODUCTS

List all pairs of students who have taken at least one class together.

```
SELECT distinct S1.name, S2.name
FROM student as S1, takes AS T1, student AS S2, takes AS T2
WHERE S1.id = T1.id AND S2.id = T2.id AND T1.course_id = T2.course_id
          AND T1.sec_id = T2.sec_id AND T1.semester = T2.semester
          AND T1.year = T2.year AND S1.id < S2.id;
```

id	*name*	*dept_name*	*tot_cred*
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

student

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

takes

inefficient use of product

what we really want:

(student × takes) AS A
with student.id = takes.id

and

(student × takes) AS B
with student.id = takes.id

PRODUCTS

```
SELECT distinct S1.name, S2.name
FROM student as S1, takes AS T1, student AS S2, takes AS T2
WHERE S1.id = T1.id AND S2.id = T2.id AND T1.course_id = T2.course_id
          AND T1.sec_id = T2.sec_id AND T1.semester = T2.semester
          AND T1.year = T2.year AND S1.id < S2.id;
```

IDEA 1 : use **nested subqueries**

```
SELECT distinct A.name, B.name
FROM (SELECT *
      FROM student, takes
      WHERE student.id = takes.id) AS A,
     (SELECT *
      FROM student, takes
      WHERE student.id = takes.id) AS B
WHERE A.course_id = B.course_id AND A.sec_id = B.sec_id
   AND A.semester = B.semester AND A.year = B.year AND A.id < B.id;
```

```
SELECT distinct S1.name, S2.name
FROM student as S1, takes AS T1, student AS S2, takes AS T2
WHERE S1.id = T1.id AND S2.id = T2.id AND T1.course_id = T2.course_id
          AND T1.sec_id = T2.sec_id AND T1.semester = T2.semester
          AND T1.year = T2.year AND S1.id < S2.id;
```

IDEA 2 : use **WITH** statement

```
WITH A AS (SELECT *
           FROM student, takes
           WHERE student.id = takes.id),
     B AS (SELECT *
           FROM student, takes
           WHERE student.id = takes.id)
SELECT distinct A.name, B.name
FROM A, B
WHERE A.course_id = B.course_id AND A.sec_id = B.sec_id
    AND A.semester = B.semester AND A.year = B.year AND A.id < B.id;
```

PRODUCTS

writing several tables in **FROM** clause automatically creates the product table

```
SELECT *
FROM student, takes
WHERE student.id = takes.id;
```

using a [,] is an implicit way to do products

common attributes appear twice in product table
student.id, takes.id

only the rows satisfying the condition survive
students who did not take any class disappear

have to specify the variables over which to 'product'
student.id = takes.id

id	*name*	*dept_name*	*tot_cred*	*id*	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	Zhang	Comp. Sci.	102	00128	CS-101	1	Fall	2009	A
00128	Zhang	Comp. Sci.	102	00128	CS-347	1	Fall	2009	A-
12345	Shanka							09	C
12345	Shanka							09	A
12345	Shanka							10	A
12345	Shanka							09	A
19991	Brandt							10	B
23121	Chavez							10	C+
44553	Peltier							09	B-
45678	Levy	Physics	46	45678	CS-101	1	Fall	2009	F
45678	Levy	Physics	46	45678	CS-101	1	Spring	2010	B+
45678	Levy	Physics	46	45678	CS-319	1	Spring	2010	B
54321	Williams	Comp. Sci.	54	54321	CS-101	1	Fall	2009	A-
54321	Williams	Comp. Sci.	54	54321	CS-190	2	Spring	2009	B+

JOIN : removes these various problems

MANY VARIATIONS

new idea

INNER JOIN

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

student

id	name	dept_name	tot_cred
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

INNER JOIN

```
SELECT *
FROM student, takes
WHERE student.id = takes.id;
```

before

id	name	dept_name	tot_cred	id	course_id	sec_id	semester	year	grade
00128	Zhang	Comp. Sci.	102	00128	CS-101	1	Fall	2009	A
00128	Zhang	Comp. Sci.	102	00128	CS-347	1	Fall	2009	A-
12345	Shankar	Comp. Sci.	32	12345	CS-101	1	Fall	2009	C
12345	Shankar	Comp. Sci.	32	12345	CS-190	2	Spring	2009	A
12345	Shankar	Comp. Sci.	32	12345	CS-315	1	Spring	2010	A
12345	Shankar	Comp. Sci.	32	12345	CS-347	1	Fall	2009	A
19991	Brandt	History	80	19991	HIS-351	1	Spring	2010	B
23121	Chavez	Finance	110	23121	FIN-201	1	Spring	2010	C+
44553	Peltier	Physics	56	44553	PHY-101	1	Fall	2009	B-
45678	Levy	Physics	46	45678	CS-101	1	Fall	2009	F
45678	Levy	Physics	46	45678	CS-101	1	Spring	2010	B+
45678	Levy	Physics	46	45678	CS-319	1	Spring	2010	B
54321	Williams	Comp. Sci.	54	54321	CS-101	1	Fall	2009	A-
54321	Williams	Comp. Sci.	54	54321	CS-190	2	Spring	2009	B+
55739	Sanchez	Music	38	55739	MU-199	1	Spring	2010	A-
76543	Brown	Comp. Sci.	58	76543	CS-101	1	Fall	2009	A
76543	Brown	Comp. Sci.	58	76543	CS-319	2	Spring	2010	A
76653	Aoi	Elec. Eng.	60	76653	EE-181	1	Spring	2009	C
98765	Bourikas	Elec. Eng.	98	98765	CS-101	1	Fall	2009	C-
98765	Bourikas	Elec. Eng.	98	98765	CS-315	1	Spring	2010	B
98988	Tanaka	Biology	120	98988	BIO-101	1	Summer	2009	A
98988	Tanaka	Biology	120	98988	BIO-301	1	Summer	2010	

new

```
SELECT *
FROM student INNER JOIN takes ON student.id = takes.id;
```

```
myuniversitydb=# SELECT *
myuniversitydb # FROM student INNER JOIN takes ON student.id = takes.id;
  id   |   name   | dept_name  | tot_cred |  id   | course_id | sec_id | semester | year | grade
-------+----------+------------+----------+-------+-----------+--------+----------+------+-------
 00128 | Zhang    | Comp. Sci. |      102 | 00128 | CS-101    | 1      | Fall     | 2009 | A
 00128 | Zhang    | Comp. Sci. |      102 | 00128 | CS-347    | 1      | Fall     | 2009 | A-
 12345 | Shankar  | Comp. Sci. |       32 | 12345 | CS-101    | 1      | Fall     | 2009 | C
 12345 | Shankar  | Comp. Sci. |       32 | 12345 | CS-190    | 2      | Spring   | 2009 | A
 12345 | Shankar  | Comp. Sci. |       32 | 12345 | CS-315    | 1      | Spring   | 2010 | A
 12345 | Shankar  | Comp. Sci. |       32 | 12345 | CS-347    | 1      | Fall     | 2009 | A
 19991 | Brandt   | History    |       80 | 19991 | HIS-351   | 1      | Spring   | 2010 | B
 23121 | Chavez   | Finance    |      110 | 23121 | FIN 201   | 1      | Spring   | 2010 | C+
 44553 | Peltier  | Physics    |       56 | 44553 | PHY-101   | 1      | Fall     | 2009 | B-
 45678 | Levy     | Physics    |       46 | 45678 | CS-101    | 1      | Fall     | 2009 | F
 45678 | Levy     | Physics    |       46 | 45678 | CS-101    | 1      | Spring   | 2010 | B+
 45678 | Levy     | Physics    |       46 | 45678 | CS-319    | 1      | Spring   | 2010 | B
 54321 | Williams | Comp. Sci. |       54 | 54321 | CS-101    | 1      | Fall     | 2009 | A-
 54321 | Williams | Comp. Sci. |       54 | 54321 | CS-190    | 2      | Spring   | 2009 | B+
 55739 | Sanchez  | Music      |       38 | 55739 | MU-199    | 1      | Spring   | 2010 | A-
 76543 | Brown    | Comp. Sci. |       58 | 76543 | CS-101    | 1      | Fall     | 2009 | A
 76543 | Brown    | Comp. Sci. |       58 | 76543 | CS-319    | 2      | Spring   | 2010 | A
 76653 | Aoi      | Elec. Eng. |       60 | 76653 | EE 181    | 1      | Spring   | 2009 | C
 98765 | Bourikas | Elec. Eng. |       98 | 98765 | CS-101    | 1      | Fall     | 2009 | C-
 98765 | Bourikas | Elec. Eng. |       98 | 98765 | CS-315    | 1      | Spring   | 2010 | B
 98988 | Tanaka   | Biology    |      120 | 98988 | BIO-101   | 1      | Summer   | 2009 | A
 98989 | Tanaka   | Biology    |      120 | 98989 | BIO-301   | 1      | Summer   | 2010 |
(22 rows)
```

Inner Join

```
SELECT *
FROM student INNER JOIN takes ON student.id = takes.id;
```

INNER JOIN

table1 INNER JOIN table2 ON **condition**

ON condition is **required**

can specify *any* condition, even if not related to the join itself

attribute id appears twice, exactly like in product

Inner Join

JOIN condition can have many parts with AND / OR

```
SELECT *
FROM student INNER JOIN takes ON student.id = takes.id AND year = 2010;
```



id	*name*	*dept_name*	*tot_cred*	*id*	*course_id*	*sec_id*	*semester*	*year*	*grade*
12345	Shankar	Comp. Sci.	32	12345	CS-315	1	Spring	2010	A
19991	Brandt	History	80	19991	HIS-351	1	Spring	2010	B
23121	Chavez	Finance	110	23121	FIN-201	1	Spring	2010	C+
45678	Levy	Physics	46	45678	CS-101	1	Spring	2010	B+
45678	Levy	Physics	46	45678	CS-319	1	Spring	2010	B
55739	Sanchez	Music	38	55739	MU-199	1	Spring	2010	A-
76543	Brown	Comp. Sci.	58	76543	CS-319	2	Spring	2010	A
98765	Bourikas	Elec. Eng.	98	98765	CS-315	1	Spring	2010	B
98988	Tanaka	Biology	120	98988	BIO-301	1	Summer	2010	

Inner Join

takes

id	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

student

id	*name*	*dept_name*	*tot_cred*
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

department

dept_name	*building*	*budget*
Biology	Watson	90000.00
Comp. Sci.	Taylor	100000.00
Elec. Eng.	Taylor	85000.00
Finance	Painter	120000.00
History	Painter	50000.00
Music	Packard	80000.00
Physics	Watson	70000.00

```
SELECT *
FROM (student INNER JOIN takes ON student.id = takes.id)
      INNER JOIN department ON student.dept_name = department.dept_name;
```



id	*name*	*dept_name*	*tot_cred*	*id*	*course_id*	*sec_id*	*semester*	*year*	*grade*	*dept_name*	*building*	*budget*
00128	Zhang	Comp. Sci.	102	00128	CS-101	1	Fall	2009	A	Comp. Sci.	Taylor	100000.00
00128	Zhang	Comp. Sci.	102	00128	CS-347	1	Fall	2009	A-	Comp. Sci.	Taylor	100000.00
12345	Shankar	Comp. Sci.	32	12345	CS-101	1	Fall	2009	C	Comp. Sci.	Taylor	100000.00
12345	Shankar	Comp. Sci.	32	12345	CS-190	2	Spring	2009	A	Comp. Sci.	Taylor	100000.00
12345	Shankar	Comp. Sci.	32	12345	CS-315	1	Spring	2010	A	Comp. Sci.	Taylor	100000.00
12345	Shankar	Comp. Sci.	32	12345	CS-347	1	Fall	2009	A	Comp. Sci.	Taylor	100000.00
19991	Brandt	History	80	19991	HIS-351	1	Spring	2010	B	History	Painter	50000.00
23121	Chavez	Finance	110	23121	FIN-201	1	Spring	2010	C+	Finance	Painter	120000.00
44553	Peltier	Physics	56	44553	PHY-101	1	Fall	2009	B-	Physics	Watson	70000.00
45678	Levy	Physics	46	45678	CS-101	1	Fall	2009	F	Physics	Watson	70000.00
45678	Levy	Physics	46	45678	CS-101	1	Spring	2010	B+	Physics	Watson	70000.00
45678	Levy	Physics	46	45678	CS-319	1	Spring	2010	B	Physics	Watson	70000.00

```
SELECT *
FROM (student INNER JOIN takes ON student.id = takes.id) AS A
          INNER JOIN department ON A.dept_name = department.dept_name;
```

id	name	dept_name	tot_cred	id	course_id	sec_id	semester	year	grade	dept_name	building	budget
00128	Zhang	Comp. Sci.	102	00128	CS-101	1	Fall	2009	A	Comp. Sci.	Taylor	100000.00
00128	Zhang	Comp. Sci.	102	00128	CS-347	1	Fall	2009	A-	Comp. Sci.	Taylor	100000.00
12345	Shankar	Comp. Sci.	32	12345	CS-101	1	Fall	2009	C	Comp. Sci.	Taylor	100000.00
12345	Shankar	Comp. Sci.	32	12345	CS-190	2	Spring	2009	A	Comp. Sci.	Taylor	100000.00
12345	Shankar	Comp. Sci.	32	12345	CS-315	1	Spring	2010	A	Comp. Sci.	Taylor	100000.00
12345	Shankar	Comp. Sci.	32	12345	CS-347	1	Fall	2009	A	Comp. Sci.	Taylor	100000.00
19991	Brandt	History	80	19991	HIS-351	1	Spring	2010	B	History	Painter	50000.00
23121	Chavez	Finance	110	23121	FIN-201	1	Spring	2010	C+	Finance	Painter	120000.00
44553	Peltier	Physics	56	44553	PHY-101	1	Fall	2009	B-	Physics	Watson	70000.00
45678	Levy	Physics	46	45678	CS-101	1	Fall	2009	F	Physics	Watson	70000.00
45678	Levy	Physics	46	45678	CS-101	1	Spring	2010	B+	Physics	Watson	70000.00
45678	Levy	Physics	46	45678	CS-319	1	Spring	2010	B	Physics	Watson	70000.00
54321	Williams	Comp. Sci.	54	54321	CS-101	1	Fall	2009	A-	Comp. Sci.	Taylor	100000.00
54321	Williams	Comp. Sci.	54	54321	CS-190	2	Spring	2009	B+	Comp. Sci.	Taylor	100000.00
55739	Sanchez	Music	38	55739	MU-199	1	Spring	2010	A-	Music	Packard	80000.00
76543	Brown	Comp. Sci.	58	76543	CS-101	1	Fall	2009	A	Comp. Sci.	Taylor	100000.00
76543	Brown	Comp. Sci.	58	76543	CS-319	2	Spring	2010	A	Comp. Sci.	Taylor	100000.00
76653	Aoi	Elec. Eng.	60	76653	EE-181	1	Spring	2009	C	Elec. Eng.	Taylor	85000.00
98765	Bourikas	Elec. Eng.	98	98765	CS-101	1	Fall	2009	C-	Elec. Eng.	Taylor	85000.00
98765	Bourikas	Elec. Eng.	98	98765	CS-315	1	Spring	2010	B	Elec. Eng.	Taylor	85000.00
98988	Tanaka	Biology	120	98988	BIO-101	1	Summer	2009	A	Biology	Watson	90000.00
98988	Tanaka	Biology	120	98988	BIO-301	1	Summer	2010		Biology	Watson	90000.00

same table as before, we just renamed the first join table

```
SELECT id
FROM (student INNER JOIN takes ON student.id = takes.id) AS A
        INNER JOIN department ON A.dept_name = department.dept_name;
```

two attributes with name id in table **A**

```
ERROR:  column reference "id" is ambiguous
LINE 1: SELECT id
               ^
```

id	name	dept_name	tot_cred	id	course_id	sec_id	semester	year	grade	dept_name	building	budget
00128	Zhang	Comp. Sci.	102	00128	CS-101	1	Fall	2009	A	Comp. Sci.	Taylor	100000.00
00128	Zhang	Comp. Sci.	102	00128	CS-347	1	Fall	2009	A-	Comp. Sci.	Taylor	100000.00
12345	Shankar	Comp. Sci.	32	12345	CS-101	1	Fall	2009	C	Comp. Sci.	Taylor	100000.00
12345	Shankar	Comp. Sci.	32	12345	CS-190	2	Spring	2009	A	Comp. Sci.	Taylor	100000.00
12345	Shankar	Comp. Sci.	32	12345	CS-315	1	Spring	2010	A	Comp. Sci.	Taylor	100000.00
12345	Shankar	Comp. Sci.	32	12345	CS-347	1	Fall	2009	A	Comp. Sci.	Taylor	100000.00
19991	Brandt	History	80	19991	HIS-351	1	Spring	2010	B	History	Painter	50000.00
23121	Chavez	Finance	110	23121	FIN-201	1	Spring	2010	C+	Finance	Painter	120000.00
44553	Peltier	Physics	56	44553	PHY-101	1	Fall	2009	B-	Physics	Watson	70000.00
45678	Levy	Physics	46	45678	CS-101	1	Fall	2009	F	Physics	Watson	70000.00
45678	Levy	Physics	46	45678	CS-101	1	Spring	2010	B+	Physics	Watson	70000.00
45678	Levy	Physics	46	45678	CS-319	1	Spring	2010	B	Physics	Watson	70000.00
54321	Williams	Comp. Sci.	54	54321	CS-101	1	Fall	2009	A-	Comp. Sci.	Taylor	100000.00
54321	Williams	Comp. Sci.	54	54321	CS-190	2	Spring	2009	B+	Comp. Sci.	Taylor	100000.00
55739	Sanchez	Music	38	55739	MU-199	1	Spring	2010	A-	Music	Packard	80000.00
76543	Brown	Comp. Sci.	58	76543	CS-101	1	Fall	2009	A	Comp. Sci.	Taylor	100000.00
76543	Brown	Comp. Sci.	58	76543	CS-319	2	Spring	2010	A	Comp. Sci.	Taylor	100000.00
76653	Aoi	Elec. Eng.	60	76653	EE-181	1	Spring	2009	C	Elec. Eng.	Taylor	85000.00
98765	Bourikas	Elec. Eng.	98	98765	CS-101	1	Fall	2009	C-	Elec. Eng.	Taylor	85000.00
98765	Bourikas	Elec. Eng.	98	98765	CS-315	1	Spring	2010	B	Elec. Eng.	Taylor	85000.00
98988	Tanaka	Biology	120	98988	BIO-101	1	Summer	2009	A	Biology	Watson	90000.00
98988	Tanaka	Biology	120	98988	BIO-301	1	Summer	2010		Biology	Watson	90000.00

Inner Join: Renaming

```
SELECT *
FROM (student INNER JOIN takes ON student.id = takes.id) AS mytable(a, b, c, d, e, f, g, h);
```

mytable

a	*b*	*c*	*d*	*e*	*f*	*g*	*h*	*year*	*grade*
00128	Zhang	Comp. Sci.	102	00128	CS-101	1	Fall	2009	A
00128	Zhang	Comp. Sci.	102	00128	CS-347	1	Fall	2009	A-
12345	Shankar	Comp. Sci.	32	12345	CS-101	1	Fall	2009	C
12345	Shankar	Comp. Sci.	32	12345	CS-190	2	Spring	2009	A
12345	Shankar	Comp. Sci.	32	12345	CS-315	1	Spring	2010	A
12345	Shankar	Comp. Sci.	32	12345	CS-347	1	Fall	2009	A
19991	Brandt	History	80	19991	HIS-351	1	Spring	2010	B
23121	Chavez	Finance	110	23121	FIN-201	1	Spring	2010	C+
44553	Peltier	Physics	56	44553	PHY-101	1	Fall	2009	B-
45678	Levy	Physics	46	45678	CS-101	1	Fall	2009	F
45678	Levy	Physics	46	45678	CS-101	1	Spring	2010	B+
45678	Levy	Physics	46	45678	CS-319	1	Spring	2010	B
54321	Williams	Comp. Sci.	54	54321	CS-101	1	Fall	2009	A-
54321	Williams	Comp. Sci.	54	54321	CS-190	2	Spring	2009	B+
55739	Sanchez	Music	38	55739	MU-199	1	Spring	2010	A-
76543	Brown	Comp. Sci.	58	76543	CS-101	1	Fall	2009	A
76543	Brown	Comp. Sci.	58	76543	CS-319	2	Spring	2010	A

```
myuniversitydb=# SELECT *
FROM (student INNER JOIN takes ON student.id = takes.id) AS mytable(a, b, c, d, e, f, g, h);
   a   |    b     |     c      |  d  |   e   |    f    | g |   h    | year | grade
-------+----------+------------+-----+-------+---------+---+--------+------+-------
 00128 | Zhang    | Comp. Sci. | 102 | 00128 | CS-101  | 1 | Fall   | 2009 | A
 00128 | Zhang    | Comp. Sci. | 102 | 00128 | CS-347  | 1 | Fall   | 2009 | A-
 12345 | Shankar  | Comp. Sci. |  32 | 12345 | CS-101  | 1 | Fall   | 2009 | C
 12345 | Shankar  | Comp. Sci. |  32 | 12345 | CS-190  | 2 | Spring | 2009 | A
 12345 | Shankar  | Comp. Sci. |  32 | 12345 | CS-315  | 1 | Spring | 2010 | A
 12345 | Shankar  | Comp. Sci. |  32 | 12345 | CS-347  | 1 | Fall   | 2009 | A
 19991 | Brandt   | History    |  80 | 19991 | HIS-351 | 1 | Spring | 2010 | B
 23121 | Chavez   | Finance    | 110 | 23121 | FIN-201 | 1 | Spring | 2010 | C+
 44553 | Peltier  | Physics    |  56 | 44553 | PHY-101 | 1 | Fall   | 2009 | B-
 45670 | Levy     | Physics    |  46 | 45670 | CS-101  | 1 | Fall   | 2009 | F
 45678 | Levy     | Physics    |  46 | 45678 | CS-101  | 1 | Spring | 2010 | B+
 45678 | Levy     | Physics    |  46 | 45678 | CS-319  | 1 | Spring | 2010 | B
 54321 | Williams | Comp. Sci. |  54 | 54321 | CS-101  | 1 | Fall   | 2009 | A-
 54321 | Williams | Comp. Sci. |  54 | 54321 | CS-190  | 2 | Spring | 2009 | B+
 55739 | Sanchez  | Music      |  38 | 55739 | MU-199  | 1 | Spring | 2010 | A-
 76543 | Brown    | Comp. Sci. |  58 | 76543 | CS-101  | 1 | Fall   | 2009 | A
 76543 | Brown    | Comp. Sci. |  58 | 76543 | CS-319  | 2 | Spring | 2010 | A
 76653 | Aoi      | Elec. Eng. |  60 | 76653 | EE-181  | 1 | Spring | 2009 | C
 98765 | Bourikas | Elec. Eng. |  98 | 98765 | CS-101  | 1 | Fall   | 2009 | C-
 98765 | Bourikas | Elec. Eng. |  98 | 98765 | CS-315  | 1 | Spring | 2010 | B
 98988 | Tanaka   | Biology    | 120 | 98988 | BIO-101 | 1 | Summer | 2009 | A
 98988 | Tanaka   | Biology    | 120 | 98988 | BIO-301 | 1 | Summer | 2010 |
(22 rows)
```

Inner Join: Renaming

can rename the new table after join

```
SELECT mytable.a
FROM (student INNER JOIN takes ON student.id = takes.id) AS mytable(a, b, c, d, e, f, g, h);
```

cannot access old names if we give the join table a name

```
SELECT student.id
FROM (student INNER JOIN takes ON student.id = takes.id) AS mytable(a, b, c, d, e, f, g, h);
```

can rename the old table before the join

```
SELECT S.id
FROM student AS S INNER JOIN takes AS T ON S.id = T.id;
```

cannot access old table name if we give the join table a name

```
SELECT S.id
FROM (student AS S INNER JOIN takes AS T ON S.id = T.id) AS A;
```

new idea

CROSS JOIN

Cross Join

```
SELECT *
FROM student CROSS JOIN takes;
```

```
SELECT *
FROM student INNER JOIN takes ON TRUE;
```

```
SELECT *
FROM student, takes;
```

new

just saw

before

id	*name*	*dept_name*	*tot_cred*	*id*	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	Zhang	Comp. Sci.	102	00128	CS-101	1	Fall	2009	A
12345	Shankar	Comp. Sci.	32	00128	CS-101	1	Fall	2009	A
19991	Brandt	History	80	00128	CS-101	1	Fall	2009	A
23121	Chavez	Finance	110	00128	CS-101	1	Fall	2009	A
44553	Peltier	Physics	56	00128	CS-101	1	Fall	2009	A
45678	Levy	Physics	46	00128	CS-101	1	Fall	2009	A
54321	Wil							009	A
55739	San							009	A
70557	Sno							009	A
76543	Bro							009	A
76653	Aoi	Elec. Eng.	60	00128	CS-101	1	Fall	2009	A
98765	Bourikas	Elec. Eng.	98	00128	CS-101	1	Fall	2009	A
98988	Tanaka	Biology	120	00128	CS-101	1	Fall	2009	A
00128	Zhang	Comp. Sci.	102	00128	CS-347	1	Fall	2009	A-
12345	Shankar	Comp. Sci.	32	00128	CS-347	1	Fall	2009	A-
19991	Brandt	History	80	00128	CS-347	1	Fall	2009	A-
23121	Chavez	Finance	110	00128	CS-347	1	Fall	2009	A-
44553	Peltier	Physics	56	00128	CS-347	1	Fall	2009	A-
45678	Levy	Physics	46	00128	CS-347	1	Fall	2009	A-
54321	Williams	Comp. Sci.	54	00128	CS-347	1	Fall	2009	A-
55739	Sanchez	Music	38	00128	CS-347	1	Fall	2009	A-

CROSS JOIN

no condition can be specified, **same as product**

new idea

OUTER JOIN

Left Outer Join

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

student

id	name	dept_name	tot_cred
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

```
SELECT *
FROM student LEFT OUTER JOIN takes
                    ON student.id = takes.id;
```

id	name	dept_name	tot_cred	id	course_id	sec_id	semester	year	grade
00128	Zhang	Comp. Sci.	102	00128	CS-101	1	Fall	2009	A
00128	Zhang	Comp. Sci.	102	00128	CS-347	1	Fall	2009	A-
12345	Shankar	Comp. Sci.	32	12345	CS-101	1	Fall	2009	C
12345	Shankar	Comp. Sci.	32	12345	CS-190	2	Spring	2009	A
12345	Shankar	Comp. Sci.	32	12345	CS-315	1	Spring	2010	A
12345	Shankar	Comp. Sci.	32	12345	CS-347	1	Fall	2009	A
19991	Brandt	History	80	19991	HIS-351	1	Spring	2010	B
23121	Chavez	Finance	110	23121	FIN-201	1	Spring	2010	C+
44553	Peltier	Physics	56	44553	PHY-101	1	Fall	2009	B-
45678	Levy	Physics	46	45678	CS-101	1	Fall	2009	F
45678	Levy	Physics	46	45678	CS-101	1	Spring	2010	B+
45678	Levy	Physics	46	45678	CS-319	1	Spring	2010	B
54321	Williams	Comp. Sci.	54	54321	CS-101	1	Fall	2009	A-
54321	Williams	Comp. Sci.	54	54321	CS-190	2	Spring	2009	B+
55739	Sanchez	Music	38	55739	MU-199	1	Spring	2010	A-
76543	Brown	Comp. Sci.	58	76543	CS-101	1	Fall	2009	A
76543	Brown	Comp. Sci.	58	76543	CS-319	2	Spring	2010	A
76653	Aoi	Elec. Eng.	60	76653	EE-181	1	Spring	2009	C
98765	Bourikas	Elec. Eng.	98	98765	CS-101	1	Fall	2009	C-
98765	Bourikas	Elec. Eng.	98	98765	CS-315	1	Spring	2010	B
98988	Tanaka	Biology	120	98988	BIO-101	1	Summer	2009	A
98988	Tanaka	Biology	120	98988	BIO-301	1	Summer	2010	
70557	Snow	Physics	0						

Left Outer Join

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

student

id	name	dept_name	tot_cred
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

```
SELECT *
FROM student LEFT OUTER JOIN takes
      ON student.id = takes.id;
```


LEFT OUTER JOIN

table1 LEFT OUTER JOIN table2 ON **condition**

— first do **INNER JOIN** to get tuples

— for each unmatched row in left table (table1)

- add row line from left table
- fill in attributes of right table as null

ON condition is **required**

Right Outer Join

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

RIGHT OUTER JOIN

table1 RIGHT OUTER JOIN table2 ON **condition**

— first do **INNER JOIN** to get tuples

— for each unmatched row in right table (table2)

- add row line from right table
- fill in attributes of left table as null

```
SELECT *
FROM student RIGHT OUTER JOIN takes
    ON student.id = takes.id;
```

id	name	dept_name	tot_cred
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

student

id	name	dept_name	tot_cred	id	course_id	sec_id	semester	year	grade
00128	Zhang	Comp. Sci.	102	00128	CS-101	1	Fall	2009	A
00128	Zhang	Comp. Sci.	102	00128	CS-347	1	Fall	2009	A-
12345	Shankar	Comp. Sci.	32	12345	CS-101	1	Fall	2009	C
12345	Shankar	Comp. Sci.	32	12345	CS-190	2	Spring	2009	A
12345	Shankar	Comp. Sci.	32	12345	CS-315	1	Spring	2010	A
12345	Shankar	Comp. Sci.	32	12345	CS-347	1	Fall	2009	A
19991	Brandt							2010	B
23121	Chavez							2010	C+
44553	Peltier							2009	B-
45678	Levy							2009	F
45678	Levy							2010	B+
45678	Levy	Physics	46	45678	CS-319	1	Spring	2010	B
54321	Williams	Comp. Sci.	54	54321	CS-101	1	Fall	2009	A-
54321	Williams	Comp. Sci.	54	54321	CS-190	2	Spring	2009	B+
55739	Sanchez	Music	38	55739	MU-199	1	Spring	2010	A-
76543	Brown	Comp. Sci.	58	76543	CS-101	1	Fall	2009	A
76543	Brown	Comp. Sci.	58	76543	CS-319	2	Spring	2010	A
76653	Aoi	Elec. Eng.	60	76653	EE-181	1	Spring	2009	C
98765	Bourikas	Elec. Eng.	98	98765	CS-101	1	Fall	2009	C-
98765	Bourikas	Elec. Eng.	98	98765	CS-315	1	Spring	2010	B
98988	Tanaka	Biology	120	98988	BIO-101	1	Summer	2009	A
98988	Tanaka	Biology	120	98988	BIO-301	1	Summer	2010	

doesn't do anything in this case, as all rows in takes are matched

Outer Join

teaches (15 rows)

id	course_id	sec_id	semester	year
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

id	name	dept_name	salary
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

teacher (12 rows)

```
SELECT *
FROM teacher RIGHT OUTER JOIN teaches
    ON teacher.id = teaches.id;
```

id	name	dept_name	salary	id	course_id	sec_id	semester	year
10101	Srinivasan	Comp. Sci.	65000.00	10101	CS-101	1	Fall	2009
10101	Srinivasan	Comp. Sci.	65000.00	10101	CS-315	1	Spring	2010
10101	Srinivasan	Comp. Sci.	65000.00	10101	CS-347	1	Fall	2009
12121	Wu	Finance	90000.00	12121	FIN-201	1	Spring	2010
15151	Mozart	Music	40000.00	15151	MU-199	1	Spring	2010
22222	Einstein	Physics	95000.00	22222	PHY-101	1	Fall	2009
32343	El Said	History	60000.00	32343	HIS-351	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	45565	CS-101	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	45565	CS-319	1	Spring	2010
76766	Crick	Biology	72000.00	76766	BIO-101	1	Summer	2009
76766	Crick	Biology	72000.00	76766	BIO-301	1	Summer	2010
83821	Brandt	Comp. Sci.	92000.00	83821	CS-190	1	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	83821	CS-190	2	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	83821	CS-319	2	Spring	2010
98345	Kim	Elec. Eng.	80000.00	98345	EE-181	1	Spring	2009

each id in teaches present in teacher

so each row in teaches matches some row in teacher

so the RIGHT OUTER JOIN contains as many rows as teaches

Outer Join

teaches (15 rows)

id	*course_id*	*sec_id*	*semester*	*year*
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

teacher (12 rows)

```
SELECT *
FROM teacher LEFT OUTER JOIN teaches
     ON teacher.id = teaches.id;
```

id	*name*	*dept_name*	*salary*	*id*	*course_id*	*sec_id*	*semester*	*year*
10101	Srinivasan	Comp. Sci.	65000.00	10101	CS-101	1	Fall	2009
10101	Srinivasan	Comp. Sci.	65000.00	10101	CS-315	1	Spring	2010
10101	Srinivasan	Comp. Sci.	65000.00	10101	CS-347	1	Fall	2009
12121	Wu	Finance	90000.00	12121	FIN-201	1	Spring	2010
15151	Mozart	Music	40000.00	15151	MU-199	1	Spring	2010
22222	Einstein	Physics	95000.00	22222	PHY-101	1	Fall	2009
32343	El Said	History	60000.00	32343	HIS-351	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	45565	CS-101	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	45565	CS-319	1	Spring	2010
76766	Crick	Biology	72000.00	76766	BIO-101	1	Summer	2009
76766	Crick	Biology	72000.00	76766	BIO-301	1	Summer	2010
83821	Brandt	Comp. Sci.	92000.00	83821	CS-190	1	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	83821	CS-190	2	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	83821	CS-319	2	Spring	2010
98345	Kim	Elec. Eng.	80000.00	98345	EE-181	1	Spring	2009
33456	Gold	Physics	87000.00					
58583	Califieri	History	62000.00					
76543	Singh	Finance	80000.00					

not every `id` in teacher **present** in teaches

so the LEFT OUTER JOIN contains as many rows as teaches PLUS the **null** rows

15 + 3 = 18

Outer Join

teaches (15 rows)

id	*course_id*	*sec_id*	*semester*	*year*
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

teacher (12 rows)

```
SELECT *
FROM teacher FULL OUTER JOIN teaches
     ON teacher.id = teaches.id;
```

id	*name*	*dept_name*	*salary*	*id*	*course_id*	*sec_id*	*semester*	*year*
10101	Srinivasan	Comp. Sci.	65000.00	10101	CS-101	1	Fall	2009
10101	Srinivasan	Comp. Sci.	65000.00	10101	CS-315	1	Spring	2010
10101	Srinivasan	Comp. Sci.	65000.00	10101	CS-347	1	Fall	2009
12121	Wu	Finance	90000.00	12121	FIN-201	1	Spring	2010
15151	Mozart	Music	40000.00	15151	MU-199	1	Spring	2010
22222	Einstein	Physics	95000.00	22222	PHY-101	1	Fall	2009
32343	El Said	History	60000.00	32343	HIS-351	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	45565	CS-101	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	45565	CS-319	1	Spring	2010
76766	Crick	Biology	72000.00	76766	BIO-101	1	Summer	2009
76766	Crick	Biology	72000.00	76766	BIO-301	1	Summer	2010
83821	Brandt	Comp. Sci.	92000.00	83821	CS-190	1	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	83821	CS-190	2	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	83821	CS-319	2	Spring	2010
98345	Kim	Elec. Eng.	80000.00	98345	EE-181	1	Spring	2009
33456	Gold	Physics	87000.00					
58583	Califieri	History	62000.00					
76543	Singh	Finance	80000.00					

FULL OUTER JOIN

LEFT OUTER JOIN + RIGHT OUTER JOIN

Outer Join

teaches (15 rows)

id	*course_id*	*sec_id*	*semester*	*year*
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

teacher (12 rows)

```
SELECT *
FROM teacher OUTER JOIN teaches
      ON teacher.id = teaches.id;
```

```
myuniversitydb=# SELECT *
myuniversitydb-# FROM teacher OUTER JOIN teaches
myuniversitydb-#    ON teacher.id = teaches.id;
ERROR:  syntax error at or near "OUTER"
LINE 2: FROM teacher OUTER JOIN teaches
                     ^
```

cannot just write OUTER JOIN

must write:

(LEFT OUTER JOIN or LEFT JOIN)

or

(RIGHT OUTER JOIN or RIGHT JOIN)

or

(FULL OUTER JOIN or FULL JOIN)

new idea

NATURAL JOIN

NATURAL JOIN

takes

```
SELECT student.id, student.name,
  student.dept_name,
  student.tot_cred, takes.course_id,
  takes.sec_id, takes.semester,
  takes.year, takes.grade
FROM student, takes
WHERE student.id = takes.id;
```

⇔

```
SELECT *
FROM student NATURAL INNER JOIN takes;
```

45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

id	*name*	*dept_name*	*tot_cred*
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

student

id	*name*	*dept_name*	*tot_cred*	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	Zhang	Comp. Sci.	102	CS-101	1	Fall	2009	A
00128	Zhang	Comp. Sci.	102	CS-347	1	Fall	2009	A-
12345	Shankar	Comp. Sci.	32	CS-101	1	Fall	2009	C
12345	Shankar	Comp. Sci.	32	CS-190	2	Spring	2009	A
12345	Shankar	Comp. Sci.	32	CS-315	1	Spring	2010	A
12345	Shankar	Comp. Sci.	32	CS-347	1	Fall	2009	A
19991	Brandt	History	80	HIS-351	1	Spring	2010	B
23121	Chavez	Finance	110	FIN-201	1	Spring	2010	C+
44553	Peltier	Physics	56	PHY-101	1	Fall	2009	B-
45678	Levy	Physics	46	CS-101	1	Fall	2009	F
45678	Levy	Physics	46	CS-101	1	Spring	2010	B+
45678	Levy	Physics	46	CS-319	1	Spring	2010	B
54321	Williams	Comp. Sci.	54	CS-101	1	Fall	2009	A-
54321	Williams	Comp. Sci.	54	CS-190	2	Spring	2009	B+
55739	Sanchez	Music	38	MU-199	1	Spring	2010	A-
76543	Brown	Comp. Sci.	58	CS-101	1	Fall	2009	A
76543	Brown	Comp. Sci.	58	CS-319	2	Spring	2010	A
76653	Aoi	Elec. Eng.	60	EE-181	1	Spring	2009	C
98765	Bourikas	Elec. Eng.	98	CS-101	1	Fall	2009	C-
98765	Bourikas	Elec. Eng.	98	CS-315	1	Spring	2010	B
98988	Tanaka	Biology	120	BIO-101	1	Summer	2009	A
98988	Tanaka	Biology	120	BIO-301	1	Summer	2010	

NATURAL keyword

automatic join ON *same name* attributes
duplicate attributes present only once
cannot choose other join attributes by adding ON

NATURAL JOIN

```
SELECT * FROM table1 (a, b, c, d, e) NATURAL INNER JOIN table2(a, g, h, i, d)
```

a	*d*	*table1.b*	*table1.c*	*table1.e*	*table2.g*	*table2.h*	*table2.i*

```
SELECT * FROM table1 (a, b, c, d, e) INNER JOIN table2(a, g, h, i, d)
                        ON table1.a = table2.a AND table1.d = table2.d
```

table1.a	*table1.d*	*table1.b*	*table1.c*	*table1.e*	*table2.a*	*table2.d*	*table2.g*	*table2.h*	*table2.i*

same table except one difference: the common attributes appear **once** with NATURAL

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

student

id	name	dept_name	tot_cred
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

```
myuniversitydb-# SELECT *
myuniversitydb # FROM student INNER JOIN takes ON student.id = takes.id;
 id    |   name   | dept_name | tot_cred |  id   | course_id | sec_id | semester | year | g
-------+----------+-----------+----------+-------+-----------+--------+----------+------+---
 00128 | Zhang    | Comp. Sci.|      102 | 00128 | CS-101    | 1      | Fall     | 2009 | A
 00128 | Zhang    | Comp. Sci.|      102 | 00128 | CS-347    | 1      | Fall     | 2009 | A
 12345 | Shankar  | Comp. Sci.|       32 | 12345 | CS-101    | 1      | Fall     | 2009 | C
 12345 | Shankar  | Comp. Sci.|       32 | 12345 | CS-190    | 2      | Spring   | 2009 | A
 12345 | Shankar  | Comp. Sci.|       32 | 12345 | CS-315    | 1      | Spring   | 2010 | A
 12345 | Shankar  | Comp. Sci.|       32 | 12345 | CS-347    | 1      | Fall     | 2009 | A
 19991 | Brandt   | History   |       80 | 19991 | HIS-351   | 1      | Spring   | 2010 | B
 23121 | Chavez   | Finance   |      110 | 23121 | FIN-201   | 1      | Spring   | 2010 | C
 44553 | Peltier  | Physics   |       56 | 44553 | PHY-101   | 1      | Fall     | 2009 | B
 45678 | Levy     | Physics   |       46 | 45678 | CS-101    | 1      | Fall     | 2009 | F
 45678 | Levy     | Physics   |       46 | 45678 | CS-101    | 1      | Spring   | 2010 | B
 45678 | Levy     | Physics   |       46 | 45678 | CS-319    | 1      | Spring   | 2010 | B
 54321 | Williams | Comp. Sci.|       54 | 54321 | CS-101    | 1      | Fall     | 2009 | A
 54321 | Williams | Comp. Sci.|       54 | 54321 | CS-190    | 2      | Spring   | 2009 | B
 55739 | Sanchez  | Music     |       38 | 55739 | MU-199    | 1      | Spring   | 2010 | A
 76543 | Brown    | Comp. Sci.|       58 | 76543 | CS-101    | 1      | Fall     | 2009 | A
 76543 | Brown    | Comp. Sci.|       58 | 76543 | CS-319    | 2      | Spring   | 2010 | A
 76653 | Aoi      | Elec. Eng.|       60 | 76653 | EE-181    | 1      | Spring   | 2009 | C
 98765 | Bourikas | Elec. Eng.|       98 | 98765 | CS-101    | 1      | Fall     | 2009 | C
 98765 | Bourikas | Elec. Eng.|       98 | 98765 | CS-315    | 1      | Spring   | 2010 | B
 98988 | Tanaka   | Biology   |      120 | 98988 | BIO-101   | 1      | Summer   | 2009 | A
 98988 | Tanaka   | Biology   |      120 | 98988 | BIO-301   | 1      | Summer   | 2010 |
(22 rows)
```

```
myuniversitydb-# SELECT *
myuniversitydb # FROM student NATURAL INNER JOIN takes;
 id    |   name   | dept_name | tot_cred | course_id | sec_id | semester | year | grade
-------+----------+-----------+----------+-----------+--------+----------+------+-------
 00128 | Zhang    | Comp. Sci.|      102 | CS-101    | 1      | Fall     | 2009 | A
 00128 | Zhang    | Comp. Sci.|      102 | CS-347    | 1      | Fall     | 2009 | A-
 12345 | Shankar  | Comp. Sci.|       32 | CS-101    | 1      | Fall     | 2009 | C
 12345 | Shankar  | Comp. Sci.|       32 | CS-190    | 2      | Spring   | 2009 | A
 12345 | Shankar  | Comp. Sci.|       32 | CS-315    | 1      | Spring   | 2010 | A
 12345 | Shankar  | Comp. Sci.|       32 | CS-347    | 1      | Fall     | 2009 | A
 19991 | Brandt   | History   |       80 | HIS-351   | 1      | Spring   | 2010 | B
 23121 | Chavez   | Finance   |      110 | FIN-201   | 1      | Spring   | 2010 | C+
 44553 | Peltier  | Physics   |       56 | PHY-101   | 1      | Fall     | 2009 | B-
 45678 | Levy     | Physics   |       46 | CS-101    | 1      | Fall     | 2009 | F
 45678 | Levy     | Physics   |       46 | CS-101    | 1      | Spring   | 2010 | B+
 45678 | Levy     | Physics   |       46 | CS-319    | 1      | Spring   | 2010 | B
 54321 | Williams | Comp. Sci.|       54 | CS-101    | 1      | Fall     | 2009 | A-
 54321 | Williams | Comp. Sci.|       54 | CS-190    | 2      | Spring   | 2009 | B+
 55739 | Sanchez  | Music     |       38 | MU-199    | 1      | Spring   | 2010 | A-
 76543 | Brown    | Comp. Sci.|       58 | CS-101    | 1      | Fall     | 2009 | A
 76543 | Brown    | Comp. Sci.|       58 | CS-319    | 2      | Spring   | 2010 | A
 76653 | Aoi      | Elec. Eng.|       60 | EE-181    | 1      | Spring   | 2009 | C
 98765 | Bourikas | Elec. Eng.|       98 | CS-101    | 1      | Fall     | 2009 | C
 98765 | Bourikas | Elec. Eng.|       98 | CS-315    | 1      | Spring   | 2010 | B
 98988 | Tanaka   | Biology   |      120 | BIO-101   | 1      | Summer   | 2009 | A
 98988 | Tanaka   | Biology   |      120 | BIO-301   | 1      | Summer   | 2010 |
(22 rows)
```

Natural Join

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

section

course_id	sec_id	semester	year	building	room_number	time_slot_id
BIO-101	1	Summer	2009	Painter	514	B
BIO-301	1	Summer	2010	Painter	514	A
CS-101	1	Fall	2009	Packard	101	H
CS-101	1	Spring	2010	Packard	101	F
CS-190	1	Spring	2009	Taylor	3128	E
CS-190	2	Spring	2009	Taylor	3128	A
CS-315	1	Spring	2010	Watson	120	D
CS-319	1	Spring	2010	Watson	100	B
CS-319	2	Spring	2010	Taylor	3128	C
CS-347	1	Fall	2009	Taylor	3128	A
EE-181	1	Spring	2009	Taylor	3128	C
FIN-201	1	Spring	2010	Packard	101	B
HIS-351	1	Spring	2010	Painter	514	C
MU-199	1	Spring	2010	Packard	101	D
PHY-101	1	Fall	2009	Watson	100	A

```
SELECT *
FROM takes NATURAL INNER JOIN section;
```

course_id	sec_id	semester	year	id	grade	building	room_number	time_slot_id
CS-101	1	Fall	2009	00128	A	Packard	101	H
CS-347	1	Fall	2009	00128	A-	Taylor	3128	A
CS-101	1	Fall	2009	12345	C	Packard	101	H
CS-190	2	Spring	2009	12345	A	Taylor	3128	A
CS-315	1	Spring	2010	12345	A	Watson	120	D
CS-347	1	Fall	2009	12345	A	Taylor	3128	A
HIS-351	1	Spring	2010	19991	B	Painter	514	C
FIN-201	1	Spring	2010	23121	C+	Packard	101	B
PHY-101	1	Fall	2009	44553	B-	Watson	100	A
CS-101	1	Fall	2009	45678	F	Packard	101	H
CS-101	1	Spring	2010	45678	B+	Packard	101	F
CS-319	1	Spring	2010	45678	B	Watson	100	B

Natural Join

teaches

id	*course_id*	*sec_id*	*semester*	*year*
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

teacher

```
SELECT *
FROM teacher NATURAL LEFT OUTER JOIN teaches;
```

id	*name*	*dept_name*	*salary*	*course_id*	*sec_id*	*semester*	*year*
10101	Srinivasan	Comp. Sci.	65000.00	CS-101	1	Fall	2009
10101	Srinivasan	Comp. Sci.	65000.00	CS-315	1	Spring	2010
10101	Srinivasan	Comp. Sci.	65000.00	CS-347	1	Fall	2009
12121	Wu	Finance	90000.00	FIN-201	1	Spring	2010
15151	Mozart	Music	40000.00	MU-199	1	Spring	2010
22222	Einstein	Physics	95000.00	PHY-101	1	Fall	2009
32343	El Said	History	60000.00	HIS-351	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	CS-101	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	CS-319	1	Spring	2010
76766	Crick	Biology	72000.00	BIO-101	1	Summer	2009
76766	Crick	Biology	72000.00	BIO-301	1	Summer	2010
83821	Brandt	Comp. Sci.	92000.00	CS-190	1	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	CS-190	2	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	CS-319	2	Spring	2010
98345	Kim	Elec. Eng.	80000.00	EE-181	1	Spring	2009
33456	Gold	Physics	87000.00				
58583	Califieri	History	62000.00				
76543	Singh	Finance	80000.00				

can use NATURAL with OUTER JOINS
just one `id` attribute present in output table

```
myuniversitydb=# SELECT *
myuniversitydb-# FROM teacher NATURAL LEFT OUTER JOIN teaches;
  id   |    name    | dept name  |  salary  | course id | sec id | semester | year
-------+------------+------------+----------+-----------+--------+----------+------
 10101 | Srinivasan | Comp. Sci. | 65000.00 | CS-101    | 1      | Fall     | 2009
 10101 | Srinivasan | Comp. Sci. | 65000.00 | CS-315    | 1      | Spring   | 2010
 10101 | Srinivasan | Comp. Sci. | 65000.00 | CS-347    | 1      | Fall     | 2009
 12121 | Wu         | Finance    | 90000.00 | FIN-201   | 1      | Spring   | 2010
 15151 | Mozart     | Music      | 40000.00 | MU-199    | 1      | Spring   | 2010
 22222 | Einstein   | Physics    | 95000.00 | PHY-101   | 1      | Fall     | 2009
 32343 | El Said    | History    | 60000.00 | HIS-351   | 1      | Spring   | 2010
 45565 | Katz       | Comp. Sci. | 75000.00 | CS-101    | 1      | Spring   | 2010
 45565 | Katz       | Comp. Sci. | 75000.00 | CS-319    | 1      | Spring   | 2010
 76766 | Crick      | Biology    | 72000.00 | BIO-101   | 1      | Summer   | 2009
 76766 | Crick      | Biology    | 72000.00 | BIO-301   | 1      | Summer   | 2010
 83821 | Brandt     | Comp. Sci. | 92000.00 | CS-190    | 1      | Spring   | 2009
 83821 | Brandt     | Comp. Sci. | 92000.00 | CS-190    | 2      | Spring   | 2009
 83821 | Brandt     | Comp. Sci. | 92000.00 | CS-319    | 2      | Spring   | 2010
 98345 | Kim        | Elec. Eng. | 80000.00 | EE-181    | 1      | Spring   | 2009
 33456 | Gold       | Physics    | 87000.00 |           |        |          |
 58583 | Califieri  | History    | 62000.00 |           |        |          |
 76543 | Singh      | Finance    | 80000.00 |           |        |          |
(18 rows)
```

Natural Join

teaches (15 rows)

id	*course_id*	*sec_id*	*semester*	*year*
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

dept_name	*building*	*budget*
Biology	Watson	90000.00
Comp. Sci.	Taylor	100000.00
Elec. Eng.	Taylor	85000.00
Finance	Painter	120000.00
History	Painter	50000.00
Music	Packard	80000.00
Physics	Watson	70000.00

department (7 rows)

```
SELECT *
FROM department NATURAL INNER JOIN teaches;
```

dept_name	*building*	*budget*	*id*	*course_id*	*sec_id*	*semester*	*year*
Biology	Watson	90000.00	10101	CS-101	1	Fall	2009
Biology	Watson	90000.00	10101	CS-315	1	Spring	2010
Biology	Watson	90000.00	10101	CS-347	1	Fall	2009
Biology	Watson	90000.00	12121	FIN-201	1	Spring	2010
Biology	Watson	90000.00	15151	MU-199	1	Spring	2010
Biology	Watson	90000.00	22222	PHY-101	1	Fall	2009
Biology	Watson	90000.00	32343	HIS-351	1	Spring	2010
Biology	Watson	90000.00	45565	CS-101	1	Spring	2010
Biology	Watson	90000.00	45565	CS-319	1	Spring	2010
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
Physics	Watson	70000.00	76766	BIO-101	1	Summer	2009
Physics	Watson	70000.00	76766	BIO-301	1	Summer	2010
Physics	Watson	70000.00	83821	CS-190	1	Spring	2009
Physics	Watson	70000.00	83821	CS-190	2	Spring	2009
Physics	Watson	70000.00	83821	CS-319	2	Spring	2010
Physics	Watson	70000.00	98345	EE-181	1	Spring	2009

$15 \times 7 = 105$

if no common attribute names, becomes like CROSS JOIN or PRODUCT

Natural Join

teaches

id	*course_id*	*sec_id*	*semester*	*year*
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

id	*name*	*dept_name*	*tot_cred*
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

student

```
SELECT *
FROM student NATURAL INNER JOIN teaches;
```

```
myuniversitydb=# SELECT *
myuniversitydb-# FROM student NATURAL INNER JOIN teaches;
 id | name | dept name | tot cred | course id | sec id | semester | year
----+------+-----------+----------+-----------+--------+----------+------
(0 rows)
```

`id` of teaches and student **not** the same!
if no matches for common attribute names, the output table is empty

Danger: NATURAL JOIN matches by **name only**, not by meaning. Here `id` is the **student** id in one table and the **teacher** id in the other.

new idea

NATURAL INNER JOINS

+

NULL

```
SELECT *
FROM student NATURAL INNER JOIN takes;
```

(22 rows)

id	*name*	*dept_name*	*tot_cred*	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	Zhang	Comp. Sci.	102	CS-101	1	Fall	2009	A
00128	Zhang	Comp. Sci.	102	CS-347	1	Fall	2009	A-
12345	Shankar	Comp. Sci.	32	CS-101	1	Fall	2009	C
12345	Shankar	Comp. Sci.	32	CS-190	2	Spring	2009	A
12345	Shankar	Comp. Sci.	32	CS-315	1	Spring	2010	A
12345	Shankar	Comp. Sci.	32	CS-347	1	Fall	2009	A
19991	Brandt	History	80	HIS-351	1	Spring	2010	B
23121	Chavez	Finance	110	FIN-201	1	Spring	2010	C+
44553							9	B-
45678							9	F
45678	Levy	Physics	46	CS-101	1	Spring	2010	B+
45678	Levy	Physics	46	CS-319	1	Spring	2010	B
54321	Williams	Comp. Sci.	54	CS-101	1	Fall	2009	A-
54321	Williams	Comp. Sci.	54	CS-190	2	Spring	2009	B+
55739	Sanchez	Music	38	MU-199	1	Spring	2010	A-
76543	Brown	Comp. Sci.	58	CS-101	1	Fall	2009	A
76543	Brown	Comp. Sci.	58	CS-319	2	Spring	2010	A
76653	Aoi	Elec. Eng.	60	EE-181	1	Spring	2009	C
98765	Bourikas	Elec. Eng.	98	CS-101	1	Fall	2009	C-
98765	Bourikas	Elec. Eng.	98	CS-315	1	Spring	2010	B
98988	Tanaka	Biology	120	BIO-101	1	Summer	2009	A
98988	Tanaka	Biology	120	BIO-301	1	Summer	2010	

NATURAL INNER JOIN around common attributes: `id`.

```
SELECT *
FROM (student NATURAL INNER JOIN takes) AS A
      NATURAL INNER JOIN
     (student NATURAL INNER JOIN takes) AS B;
```

(21 rows)

id	*name*	*dept_name*	*tot_cred*	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	Zhang	Comp. Sci.	102	CS-101	1	Fall	2009	A
00128	Zhang	Comp. Sci.	102	CS-347	1	Fall	2009	A-
12345	Shankar	Comp. Sci.	32	CS-101	1	Fall	2009	C
12345	Shankar	Comp. Sci.	32	CS-190	2	Spring	2009	A
12345	Shankar	Comp. Sci.	32	CS-315	1	Spring	2010	A
12345	Shankar	Comp. Sci.	32	CS-347	1	Fall	2009	A
19991	Brandt	History	80	HIS-351	1	Spring	2010	B

NATURAL INNER JOIN around common attributes:
`id, name, dept_name, tot_cred, course_id, sec_id, semester, year, grade.`

45678	Levy	Physics	46	CS-101	1	Fall	2009	F
45678	Levy	Physics	46	CS-101	1	Spring	2010	B+
45678	Levy	Physics	46	CS-319	1	Spring	2010	B
54321	Williams	Comp. Sci.	54	CS-101	1	Fall	2009	A-
54321	Williams	Comp. Sci.	54	CS-190	2	Spring	2009	B+
55739	Sanchez	Music	38	MU-199	1	Spring	2010	A-
76543	Brown	Comp. Sci.	58	CS-101	1	Fall	2009	A
76543	Brown	Comp. Sci.	58	CS-319	2	Spring	2010	A
76653	Aoi	Elec. Eng.	60	EE-181	1	Spring	2009	C
98765	Bourikas	Elec. Eng.	98	CS-101	1	Fall	2009	C-
98765	Bourikas	Elec. Eng.	98	CS-315	1	Spring	2010	B
98988	Tanaka	Biology	120	BIO-101	1	Summer	2009	A

```
SELECT *
FROM (student NATURAL INNER JOIN takes) AS A
      NATURAL INNER JOIN
     (student NATURAL INNER JOIN takes) AS B;
```

(21 rows)

id	*name*	*dept_name*	*tot_cred*	*course_id*	*sec_id*	*semester*	*year*	*grade*
00128	Zhang	Comp. Sci.	102	CS-101	1	Fall	2009	A
00128	Zhang	Comp. Sci.	102	CS-347	1	Fall	2009	A-
12345	Shankar	Comp. Sci.	32	CS-101	1	Fall	2009	C
12345	Shankar	Comp. Sci.	32	CS-190	2	Spring	2009	A
12345	Shankar	Comp. Sci.	32	CS-315	1	Spring	2010	A
12345	Shankar	Comp. Sci.	32	CS-347	1	Fall	2009	A
19991	B							
23121	C							C+
44553	P							B-
45678	L							
45678	L							B+
45678	Levy	Physics	46	CS-319	1	Spring	2010	B
54321	Williams	Comp. Sci.	54	CS-101	1	Fall	2009	A-
54321	Williams	Comp. Sci.	54	CS-190	2	Spring	2009	B+
55739	Sanchez	Music	38	MU-199	1	Spring	2010	A-
76543	Brown	Comp. Sci.	58	CS-101	1	Fall	2009	A
76543	Brown	Comp. Sci.	58	CS-319	2	Spring	2010	A
76653	Aoi	Elec. Eng.	60	EE-181	1	Spring	2009	C
98765	Bourikas	Elec. Eng.	98	CS-101	1	Fall	2009	C-
98765	Bourikas	Elec. Eng.	98	CS-315	1	Spring	2010	B
98988	Tanaka	Biology	120	BIO-101	1	Summer	2009	A

Something weird: this table has: **21** rows !

Why the difference from previous table with **22** rows:

null values equality testing returns **false**!

last line with `Tanaka` is missing due to null grade.

```
SELECT *
FROM student NATURAL INNER JOIN takes;
```

(22 rows)

id	name	dept_name	tot_cred	course_id	sec_id	semester	year	grade
00128	Zhang	Comp. Sci.	102	CS-101	1	Fall	2009	A
00128	Zhang	Comp. Sci.	102	CS-347	1	Fall	2009	A-
12345	Shankar	Comp. Sci.	32	CS-101	1	Fall	2009	C
12345	Shankar	Comp. Sci.	32	CS-190	2	Spring	2009	A
12345	Shankar	Comp. Sci.	32	CS-315	1	Spring	2010	A
12345	Shankar	Comp. Sci.	32	CS-347	1	Fall	2009	A
19991	Brandt	History	80	HIS-351	1	Spring	2010	B
23121	Chavez	Finance	110	FIN-201	1	Spring	2010	C+
44553	Peltier	Physics	56	PHY-101	1	Fall	2009	B-
45678	Levy	Physics	46	CS-101	1	Fall	2009	F
45678	Levy	Physics	46	CS-101	1	Spring	2010	B+
45678	Levy	Physics	46	CS-319	1	Spring	2010	B
54321	Williams	Comp. Sci.	54	CS-101	1	Fall	2009	A-
54321	Williams	Comp. Sci.	54	CS-190	2	Spring	2009	B+
55739	Sanchez	Music	38	MU-199	1	Spring	2010	A-
76543	Brown	Comp. Sci.	58	CS-101	1	Fall	2009	A
76543	Brown	Comp. Sci.	58	CS-319	2	Spring	2010	A
76653	Aoi	Elec. Eng.	60	EE-181	1	Spring	2009	C
98765	Bourikas	Elec. Eng.	98	CS-101	1	Fall	2009	C-
98765	Bourikas	Elec. Eng.	98	CS-315	1	Spring	2010	B
98988	Tanaka	Biology	120	BIO-101	1	Summer	2009	A
98988	Tanaka	Biology	120	BIO-301	1	Summer	2010	

```
SELECT *
FROM (student NATURAL INNER JOIN takes) AS A
      NATURAL INNER JOIN
     (student NATURAL INNER JOIN takes) AS B;
```

(21 rows)

id	name	dept_name	tot_cred	course_id	sec_id	semester	year	grade
00128	Zhang	Comp. Sci.	102	CS-101	1	Fall	2009	A
00128	Zhang	Comp. Sci.	102	CS-347	1	Fall	2009	A-
12345	Shankar	Comp. Sci.	32	CS-101	1	Fall	2009	C
12345	Shankar	Comp. Sci.	32	CS-190	2	Spring	2009	A
12345	Shankar	Comp. Sci.	32	CS-315	1	Spring	2010	A
12345	Shankar	Comp. Sci.	32	CS-347	1	Fall	2009	A
19991	Brandt	History	80	HIS-351	1	Spring	2010	B
23121	Chavez	Finance	110	FIN-201	1	Spring	2010	C+
44553	Peltier	Physics	56	PHY-101	1	Fall	2009	B-
45678	Levy	Physics	46	CS-101	1	Fall	2009	F
45678	Levy	Physics	46	CS-101	1	Spring	2010	B+
45678	Levy	Physics	46	CS-319	1	Spring	2010	B
54321	Williams	Comp. Sci.	54	CS-101	1	Fall	2009	A-
54321	Williams	Comp. Sci.	54	CS-190	2	Spring	2009	B+
55739	Sanchez	Music	38	MU-199	1	Spring	2010	A-
76543	Brown	Comp. Sci.	58	CS-101	1	Fall	2009	A
76543	Brown	Comp. Sci.	58	CS-319	2	Spring	2010	A
76653	Aoi	Elec. Eng.	60	EE-181	1	Spring	2009	C
98765	Bourikas	Elec. Eng.	98	CS-101	1	Fall	2009	C-
98765	Bourikas	Elec. Eng.	98	CS-315	1	Spring	2010	B
98988	Tanaka	Biology	120	BIO-101	1	Summer	2009	A

Something weird: right table has: **21** rows !

Why the difference from left table with **22** rows:

null values equality testing returns **false**!

last line with **Tanaka** is missing due to null grade.

new idea

USING

Joins with USING

Often same attributes have the same name across tables

teacher.id teaches.id

parenthesis required

In this case, use simpler syntax **USING** to do a join

```
SELECT *
FROM teacher INNER JOIN department USING (dept_name)
```

⇕

```
SELECT *
FROM teacher INNER JOIN department ON
        teacher.dept_name = department.dept_name
```

one difference: the common attributes appear **once** for USING

```
SELECT * FROM table1 (a,b,c) INNER JOIN table2(a,b,d) USING (a)
```

a	*table1.b*	*table1.c*	*table2.b*	*table2.d*

```
SELECT *
FROM teacher INNER JOIN department USING (dept_name);
```

dept_name	*id*	*name*	*salary*	*building*	*budget*
Comp. Sci.	10101	Srinivasan	65000.00	Taylor	100000.00
Finance	12121	Wu	90000.00	Painter	120000.00
Music	15151	Mozart	40000.00	Packard	80000.00
Physics	22222	Einstein	95000.00	Watson	70000.00
History	32343	El Said	60000.00	Painter	50000.00
Physics	33456	Gold	87000.00	Watson	70000.00
Comp. Sci.	45565	Katz	75000.00	Taylor	100000.00
History	58583	Califieri	62000.00	Painter	50000.00
Finance	76543	Singh	80000.00	Painter	120000.00
Biology	76766	Crick	72000.00	Watson	90000.00
Comp. Sci.	83821	Brandt	92000.00	Taylor	100000.00
Elec. Eng.	98345	Kim	80000.00	Taylor	85000.00

```
SELECT *
FROM teacher INNER JOIN department
     ON teacher.dept_name = department.dept_name;
```

id	*name*	*dept_name*	*salary*	*dept_name*	*building*	*budget*
10101	Srinivasan	Comp. Sci.	65000.00	Comp. Sci.	Taylor	100000.00
12121	Wu	Finance	90000.00	Finance	Painter	120000.00
15151	Mozart	Music	40000.00	Music	Packard	80000.00
22222	Einstein	Physics	95000.00	Physics	Watson	70000.00
32343	El Said	History	60000.00	History	Painter	50000.00
33456	Gold	Physics	87000.00	Physics	Watson	70000.00
45565	Katz	Comp. Sci.	75000.00	Comp. Sci.	Taylor	100000.00
58583	Califieri	History	62000.00	History	Painter	50000.00
76543	Singh	Finance	80000.00	Finance	Painter	120000.00
76766	Crick	Biology	72000.00	Biology	Watson	90000.00
83821	Brandt	Comp. Sci.	92000.00	Comp. Sci.	Taylor	100000.00
98345	Kim	Elec. Eng.	80000.00	Elec. Eng.	Taylor	85000.00

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

section

course_id	sec_id	semester	year	building	room_number	time_slot_id
BIO-101	1	Summer	2009	Painter	514	B
BIO-301	1	Summer	2010	Painter	514	A
CS-101	1	Fall	2009	Packard	101	H
CS-101	1	Spring	2010	Packard	101	F
CS-190	1	Spring	2009	Taylor	3128	E
CS-190	2	Spring	2009	Taylor	3128	A
CS-315	1	Spring	2010	Watson	120	D
CS-319	1	Spring	2010	Watson	100	B
CS-319	2	Spring	2010	Taylor	3128	C
CS-347	1	Fall	2009	Taylor	3128	A
EE-181	1	Spring	2009	Taylor	3128	C
FIN-201	1	Spring	2010	Packard	101	B
HIS-351	1	Spring	2010	Painter	514	C
MU-199	1	Spring	2010	Packard	101	D
PHY-101	1	Fall	2009	Watson	100	A

```
myuniversitydb=# SELECT *
myuniversitydb-# FROM takes INNER JOIN section USING (course_id, sec_id, semester, year);
 course_id | sec_id | semester | year |   id   | grade | building | room_number | time_slot_id
-----------+--------+----------+------+--------+-------+----------+-------------+--------------
 CS-101    | 1      | Fall     | 2009 | 00128  | A     | Packard  | 101         | H
 CS-347    | 1      | Fall     | 2009 | 00128  | A-    | Taylor   | 3128        | A
 CS-101    | 1      | Fall     | 2009 | 12345  | C     | Packard  | 101         | H
 CS-190    | 2      | Spring   | 2009 | 12345  | A     | Taylor   | 3128        | A
 CS-315    | 1      | Spring   | 2010 | 12345  | A     | Watson   | 120         | D
 CS-347    | 1      | Fall     | 2009 | 12345  | A     | Taylor   | 3128        | A
 HIS-351   | 1      | Spring   | 2010 | 19991  | B     | Painter  | 514         | C
 FIN-201   | 1      | Spring   | 2010 | 23121  | C+    | Packard  | 101         | B
 PHY-101   | 1      | Fall     | 2009 | 44553  | B-    | Watson   | 100         | A
 CS-101    | 1      | Fall     | 2009 | 45678  | F     | Packard  | 101         | H
 CS-101    | 1      | Spring   | 2010 | 45678  | B+    | Packard  | 101         | F
 CS-319    | 1      | Spring   | 2010 | 45678  | B     | Watson   | 100         | B
 CS-101    | 1      | Fall     | 2009 | 54321  | A-    | Packard  | 101         | H
 CS-190    | 2      | Spring   | 2009 | 54321  | B+    | Taylor   | 3128        | A
 MU-199    | 1      | Spring   | 2010 | 55739  | A-    | Packard  | 101         | D
 CS-101    | 1      | Fall     | 2009 | 76543  | A     | Packard  | 101         | H
 CS-319    | 2      | Spring   | 2010 | 76543  | A     | Taylor   | 3128        | C
 EE-181    | 1      | Spring   | 2009 | 76653  | C     | Taylor   | 3128        | C
 CS-101    | 1      | Fall     | 2009 | 98765  | C     | Packard  | 101         | H
 CS-315    | 1      | Spring   | 2010 | 98765  | B     | Watson   | 120         | D
 BIO-101   | 1      | Summer   | 2009 | 98988  | A     | Painter  | 514         | B
 BIO-301   | 1      | Summer   | 2010 | 98988  |       | Painter  | 514         | A
(22 rows)
```

attributes in USING must belong to both tables

```
SELECT *
FROM takes INNER JOIN section USING (id);
```

```
myuniversitydb=# SELECT *
myuniversitydb-# FROM takes INNER JOIN section USING (id);
ERROR:  column "id" specified in USING clause does not exist in right table
```

takes

id	course_id	sec_id	semester	year	grade
00128	CS-101	1	Fall	2009	A
00128	CS-347	1	Fall	2009	A-
12345	CS-101	1	Fall	2009	C
12345	CS-190	2	Spring	2009	A
12345	CS-315	1	Spring	2010	A
12345	CS-347	1	Fall	2009	A
19991	HIS-351	1	Spring	2010	B
23121	FIN-201	1	Spring	2010	C+
44553	PHY-101	1	Fall	2009	B-
45678	CS-101	1	Fall	2009	F
45678	CS-101	1	Spring	2010	B+
45678	CS-319	1	Spring	2010	B
54321	CS-101	1	Fall	2009	A-
54321	CS-190	2	Spring	2009	B+
55739	MU-199	1	Spring	2010	A-
76543	CS-101	1	Fall	2009	A
76543	CS-319	2	Spring	2010	A
76653	EE-181	1	Spring	2009	C
98765	CS-101	1	Fall	2009	C-
98765	CS-315	1	Spring	2010	B
98988	BIO-101	1	Summer	2009	A
98988	BIO-301	1	Summer	2010	

section

course_id	sec_id	semester	year	building	room_number	time_slot_id
BIO-101	1	Summer	2009	Painter	514	B
BIO-301	1	Summer	2010	Painter	514	A
CS-101	1	Fall	2009	Packard	101	H
CS-101	1	Spring	2010	Packard	101	F
CS-190	1	Spring	2009	Taylor	3128	E
CS-190	2	Spring	2009	Taylor	3128	A
CS-315	1	Spring	2010	Watson	120	D
CS-319	1	Spring	2010	Watson	100	B
CS-319	2	Spring	2010	Taylor	3128	C
CS-347	1	Fall	2009	Taylor	3128	A
EE-181	1	Spring	2009	Taylor	3128	C
FIN-201	1	Spring	2010	Packard	101	B
HIS-351	1	Spring	2010	Painter	514	C
MU-199	1	Spring	2010	Packard	101	D
PHY-101	1	Fall	2009	Watson	100	A

JOINS are executed from left to right



```
SELECT *
FROM teacher NATURAL INNER JOIN teaches NATURAL INNER JOIN department;
```



```
SELECT *
FROM (teacher NATURAL INNER JOIN teaches) NATURAL INNER JOIN department;
```

JOINS: SUMMARY

JOIN is the same as INNER JOIN

some things to keep in mind

LEFT JOIN is the same as LEFT OUTER JOIN

RIGHT JOIN is the same as RIGHT OUTER JOIN

NATURAL JOIN is the same as NATURAL INNER JOIN

must specify LEFT or RIGHT or FULL before OUTER JOIN

must specify ON or USING with INNER and OUTER JOINS

must not specify ON or USING or NATURAL with CROSS JOIN

can use NATURAL before INNER JOIN or LEFT OUTER JOIN or RIGHT OUTER JOIN or FULL JOIN

advisor

s_id	*i_id*
00128	45565
12345	10101
23121	76543
44553	22222
45678	22222
76543	45565
76653	98345
98765	98345
98988	76766

the order of execution of JOINS is important!

List all students and their advisors, including students who don't have an assigned advisor.

```
SELECT student.name, teacher.name
FROM (student LEFT JOIN advisor ON student.id = s_id)
        INNER JOIN teacher ON i_id = teacher.id;
```

student

id	*name*	*dept_name*	*tot_cred*
00128	Zhang	Comp. Sci.	102
12345	Shankar	Comp. Sci.	32
19991	Brandt	History	80
23121	Chavez	Finance	110
44553	Peltier	Physics	56
45678	Levy	Physics	46
54321	Williams	Comp. Sci.	54
55739	Sanchez	Music	38
70557	Snow	Physics	0
76543	Brown	Comp. Sci.	58
76653	Aoi	Elec. Eng.	60
98765	Bourikas	Elec. Eng.	98
98988	Tanaka	Biology	120

name	*name*
Zhang	Katz
Shankar	Srinivasan
Chavez	Singh
Peltier	Einstein
Levy	Einstein
Brown	Katz
Aoi	Kim
Bourikas	Kim
Tanaka	Crick

correct: INNER JOIN first, then LEFT JOIN keeps **all** students

name	*name*
Zhang	Katz
Shankar	Srinivasan
Chavez	Singh
Peltier	Einstein
Levy	Einstein
Brown	Katz
Aoi	Kim
Bourikas	Kim
Tanaka	Crick
Snow	
Brandt	
Williams	
Sanchez	

teacher

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

wrong: the later INNER JOIN **drops** students without an advisor

```
SELECT student.name, teacher.name
FROM student LEFT JOIN
    (advisor INNER JOIN teacher ON i_id = teacher.id)
    ON student.id = s_id;
```

new idea

JOINS with AGGREGATIONS

Joins and Aggregate Functions

```
SELECT *
FROM teacher LEFT OUTER JOIN teaches
     ON teacher.id = teaches.id;
```

teaches (15 rows)

id	course_id	sec_id	semester	year
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

(18 rows)

id	name	dept_name	salary	id	course_id	sec_id	semester	year
10101	Srinivasan	Comp. Sci.	65000.00	10101	CS-101	1	Fall	2009
10101	Srinivasan	Comp. Sci.	65000.00	10101	CS-315	1	Spring	2010
10101	Srinivasan	Comp. Sci.	65000.00	10101	CS-347	1	Fall	2009
12121	Wu	Finance	90000.00	12121	FIN-201	1	Spring	2010
15151	Mozart	Music	40000.00	15151	MU-199	1	Spring	2010
22222	Einstein	Physics	95000.00	22222	PHY-101	1	Fall	2009
32343	El Said	History	60000.00	32343	HIS-351	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	45565	CS-101	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	45565	CS-319	1	Spring	2010
76766	Crick	Biology	72000.00	76766	BIO-101	1	Summer	2009
76766	Crick	Biology	72000.00	76766	BIO-301	1	Summer	2010
83821	Brandt	Comp. Sci.	92000.00	83821	CS-190	1	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	83821	CS-190	2	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	83821	CS-319	2	Spring	2010
98345	Kim	Elec. Eng.	80000.00	98345	EE-181	1	Spring	2009
33456	Gold	Physics	87000.00					
58583	Califieri	History	62000.00					
76543	Singh	Finance	80000.00					

teacher (12 rows)

id	name	dept_name	salary
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

Recall: aggregate functions do not take into account null

SELECT count(*) — **18**
FROM teacher LEFT OUTER JOIN teaches ON teacher.id = teaches.id

SELECT count(teacher.id) — **18**
FROM teacher LEFT OUTER JOIN teaches ON teacher.id = teaches.id

SELECT count(teaches.id) — **15**
FROM teacher LEFT OUTER JOIN teaches ON teacher.id = teaches.id

Example

teacher

id	name	dept_name	salary
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

Find the number of courses taught by each teacher.

```
SELECT teacher.name, teacher.id, count(*)
FROM teacher INNER JOIN teaches
       ON teacher.id = teaches.id
GROUP BY teacher.name, teacher.id
ORDER BY count(*) DESC;
```

id	course_id	sec_id	semester	year
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

teaches

name	id	count
Brandt	83821	3
Srinivasan	10101	3
Katz	45565	2
Crick	76766	2
Kim	98345	1
Mozart	15151	1
El Said	32343	1
Wu	12121	1
Einstein	22222	1

Incorrect: missing teachers that did not teach!

Example

teacher

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

Find the number of courses taught by each teacher.

```
SELECT teacher.name, teacher.id, count(*)
FROM teacher LEFT OUTER JOIN teaches USING (id)
GROUP BY teacher.name, teacher.id
ORDER BY count(*) DESC;
```

id	*course_id*	*sec_id*	*semester*	*year*
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

teaches

name	*id*	*count*
Srinivasan	10101	3
Brandt	83821	3
Katz	45565	2
Crick	76766	2
Einstein	22222	1
Singh	76543	1
Wu	12121	1
Kim	98345	1
Mozart	15151	1
El Said	32343	1
Gold	33456	1
Califieri	58583	1

SELECT *
FROM teacher LEFT OUTER JOIN teaches USING (id)

id	*name*	*dept_name*	*salary*	*course_id*	*sec_id*	*semester*	*year*
10101	Srinivasan	Comp. Sci.	65000.00	CS-101	1	Fall	2009
10101	Srinivasan	Comp. Sci.	65000.00	CS-315	1	Spring	2010
10101	Srinivasan	Comp. Sci.	65000.00	CS-347	1	Fall	2009
12121	Wu	Finance	90000.00	FIN-201	1	Spring	2010
15151	Mozart	Music	40000.00	MU-199	1	Spring	2010
22222	Einstein	Physics	95000.00	PHY-101	1	Fall	2009
32343	El Said	History	60000.00	HIS-351	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	CS-101	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	CS-319	1	Spring	2010
76766	Crick	Biology	72000.00	BIO-101	1	Summer	2009
76766	Crick	Biology	72000.00	BIO-301	1	Summer	2010
83821	Brandt	Comp. Sci.	92000.00	CS-190	1	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	CS-190	2	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	CS-319	2	Spring	2010
98345	Kim	Elec. Eng.	80000.00	EE-181	1	Spring	2009
33456	Gold	Physics	87000.00				
58583	Califieri	History	62000.00				
76543	Singh	Finance	80000.00				

Incorrect: teachers that did not teach have 1!

Example

teacher

id	*name*	*dept_name*	*salary*
10101	Srinivasan	Comp. Sci.	65000.00
12121	Wu	Finance	90000.00
15151	Mozart	Music	40000.00
22222	Einstein	Physics	95000.00
32343	El Said	History	60000.00
33456	Gold	Physics	87000.00
45565	Katz	Comp. Sci.	75000.00
58583	Califieri	History	62000.00
76543	Singh	Finance	80000.00
76766	Crick	Biology	72000.00
83821	Brandt	Comp. Sci.	92000.00
98345	Kim	Elec. Eng.	80000.00

Find the number of courses taught by each teacher.

```
SELECT teacher.name, teacher.id, count(course_id)
FROM teacher LEFT OUTER JOIN teaches USING (id)
GROUP BY teacher.name, teacher.id
ORDER BY count(course_id) DESC;
```

id	*course_id*	*sec_id*	*semester*	*year*
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

teaches

name	*id*	*count*
Srinivasan	10101	3
Brandt	83821	3
Katz	45565	2
Crick	76766	2
Einstein	22222	1
Kim	98345	1
Mozart	15151	1
El Said	32343	1
Wu	12121	1
Singh	76543	0
Gold	33456	0
Califieri	58583	0

SELECT *
FROM teacher LEFT OUTER JOIN teaches USING (id)

id	*name*	*dept_name*	*salary*	*course_id*	*sec_id*	*semester*	*year*
10101	Srinivasan	Comp. Sci.	65000.00	CS-101	1	Fall	2009
10101	Srinivasan	Comp. Sci.	65000.00	CS-315	1	Spring	2010
10101	Srinivasan	Comp. Sci.	65000.00	CS-347	1	Fall	2009
12121	Wu	Finance	90000.00	FIN-201	1	Spring	2010
15151	Mozart	Music	40000.00	MU-199	1	Spring	2010
22222	Einstein	Physics	95000.00	PHY-101	1	Fall	2009
32343	El Said	History	60000.00	HIS-351	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	CS-101	1	Spring	2010
45565	Katz	Comp. Sci.	75000.00	CS-319	1	Spring	2010
76766	Crick	Biology	72000.00	BIO-101	1	Summer	2009
76766	Crick	Biology	72000.00	BIO-301	1	Summer	2010
83821	Brandt	Comp. Sci.	92000.00	CS-190	1	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	CS-190	2	Spring	2009
83821	Brandt	Comp. Sci.	92000.00	CS-319	2	Spring	2010
98345	Kim	Elec. Eng.	80000.00	EE-181	1	Spring	2009
33456	Gold	Physics	87000.00				
58583	Califieri	History	62000.00				
76543	Singh	Finance	80000.00				

Logical Order of Processing

